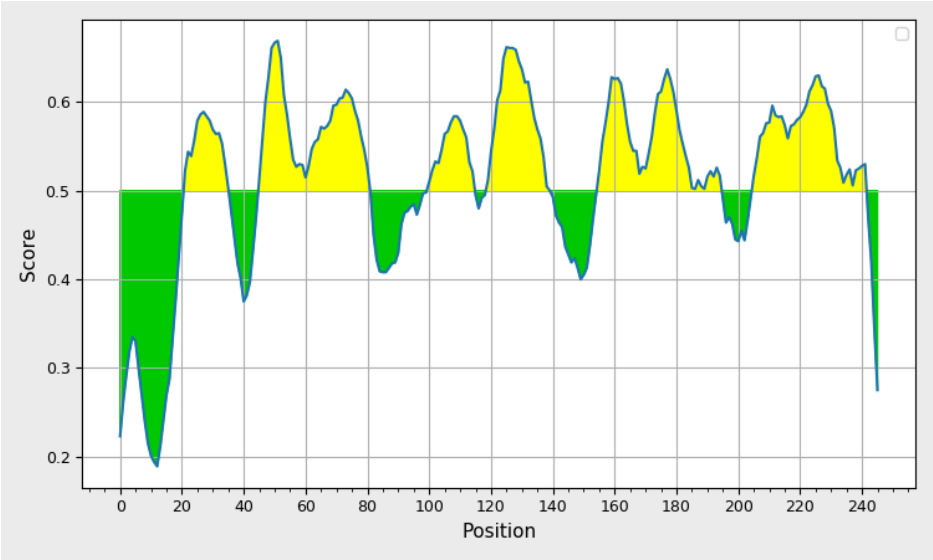
B-Cell-Epitope Prediction using BEPIPERED LINEAR EPITOPE PREDICTION 2.0 Results
(Antigenecity Score - >0.5 : probable ANTIGEN ; <0.5 : NON-ANTIGEN)

Protein : 1
Input Sequence : MPXV|URK20440.1|Chemokine binding protein - Probable ANTIGEN with B-Cell Epiopes.
>MKQYIVLACMCLVAAAMPTSLQQSSSSCTEEENKHHMGIDVIIK VTKQDQTPTNDKICQSVTEVTETEDDEVSEEVVKGDPTTYYTIVGAGLNMNFGFTKCP KILSISESSDGNTVNTRLSSVSPGQGKDSPAITREEALAMIKDCEMSIDIRCSEEEKD SDIKTHPVLGSNISHKKVSYKDIIGSTIVDTKCVKNLEFSVRIGDMCEESSELEVKDG FKYVDGSASEGATDDTSLIDSTKLKACV

Predicted peptides & Score:

No.	Start	End	Peptide	Length
1	22	36	QQSSSSCTEEENKHH	15
2	46	81	TKQDQTPTNDKICQSVTEVTETEDDEVSEEVVKGDP	36
3	101	115	CPKILSISESSDGNT	15
4	120	140	LSSVSPGQGKDSPAITREEAL	21
5	156	195	EEEKDSDIKTHPVLGSNISHKKVSYKDIIGSTIVDTKCVK	40
6	206	242	MCEESSELEVKDGFKYVDGSASEGATDDTSLIDSTKL	37



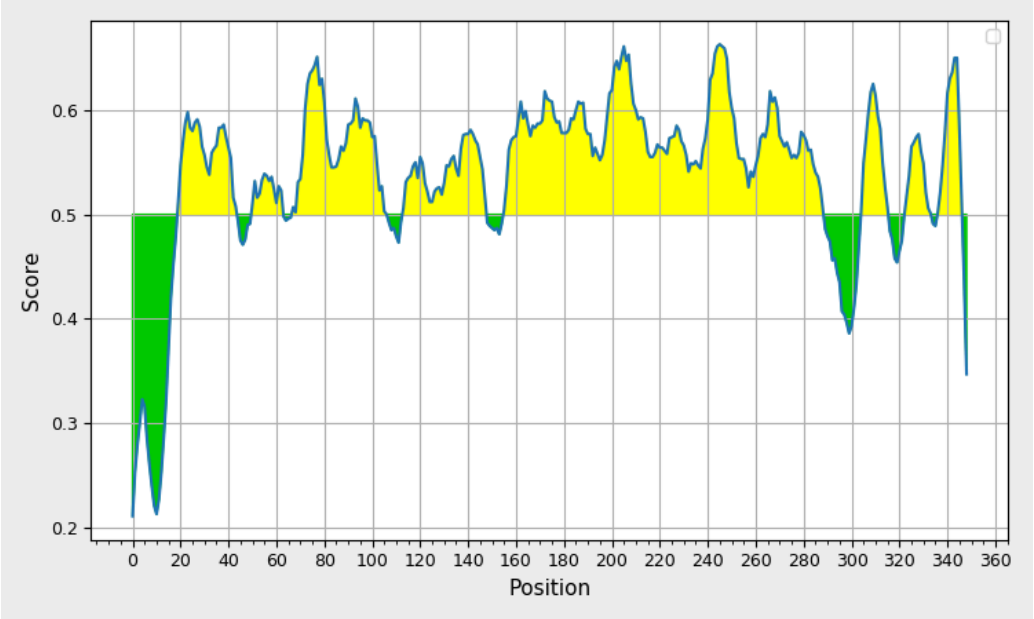
Protein : 2

Input Sequence : MPXV|URK20441.1|J2L TNF receptor (CrmB) - Probable ANTIGEN with B-Cell Epitopes

>MRSVLYSYILFLSCIIINGRDIAPHAPSNGKCKDNEYRSRNLCC LSCPPGTYAFRLCDSKTNTQCTPCGSDTFTSHNNHLQACLSCNGRCDSNQVETRSCNT THNRICECSPGYYCLLKGASGCRTCISKTKCGIGYGVSGYTSTGDVICSPCGPGTYSH TVSSTDKCEPVVTSNTFNYIDVEINLYPVNDTSCTRTTTTGLSESISTSELTITMNHK DCDPVFRAEYFSVLNNVATSGFFTGENRYQNTSKICTLNFEIKCNNKDSSSKQLTKTK NDTIMPHSETVTLVGDCLSSVDIYILYSNTNTQDYENDTISYHMGNVLDVNSHMPASC DIHKLITNSQNPTHL

Predicted peptides & Score:

No.	Start	End	Peptide	Length
1	20	44	RDIAPHAPSNGKCKDNEYRSRNLCC	25
2	51	63	TYAFRLCDSKTNT	13
3	68	106	CGSDTFTSHNNHLQACLSCNGRCDSNQVETRSCNTTHNR	39
4	114	148	YYCLLKGASGCRTCISKTKCGIGYGVSGYTSTGDV	35
5	156	289	GTYSHTVSSTDKCEPVVTSNTFNYIDVEINLYPVNDTSCTRTTTTGLSESISTSELTITMNHK DCDPVFRAEYFSVLNNVATSGFFTGENRYQNTSKICTLNFEIKCNNKDSSSKQLTKTKNDTIM PHSETVTL	134
6	305	316	NTNTQDYENDTI	12
7	324	334	LDVNSHMPASC	11
8	337	347	HKLITNSQNPT	11



Protein : 3

Input Sequences : MPXV|URK20442.1|Ankyrin - Probable ANTIGEN with B-Cell Epitopes

>MDEMDEIVRIVNDSMWYVPNAFMDDGDNEGHISVNNVCHMYLAF

FDVDISSHLFKLVIKHCDLNKRLKCGNSPLHCYTMNTRFNPSVLKILLRHGMRNFDSKDKKGHIPLHHYLIHSLSI

DNKIFDILTDPIDDFSKSSDLLLCYLRYKFNGSLNYYVLYKLLTKGSDPNCVDEDGLTSLHYYCKHISAFHESNYYK

SKSHTKMRAEKRFIYAIIDHGANINAVTKIGNTPLHTYLQQYTKHSPRVVYALLSRGADTRIRNNLNCTPIMEYIKN

DCATGHILIMLLNWHEQKYGKLQKEEGQHLLYLFIKHNQGYGSRSLNILRYLLDRFDIQKDEYYNTMTPLHTAF

QNCNNNVASYLVYIGYDINLPTKDDKTVFDLVFENRNIIYKADVVNDIIHHRLKVSLPMIKSLFYKMSEFSPYDDHY

VKKIIAYCLLRDESFAELHTKFCLNEDYKSVFMKNISFDKIDSIIEKCSRDISLLKEIRISDTDLYTVLRTEDIRYHTY

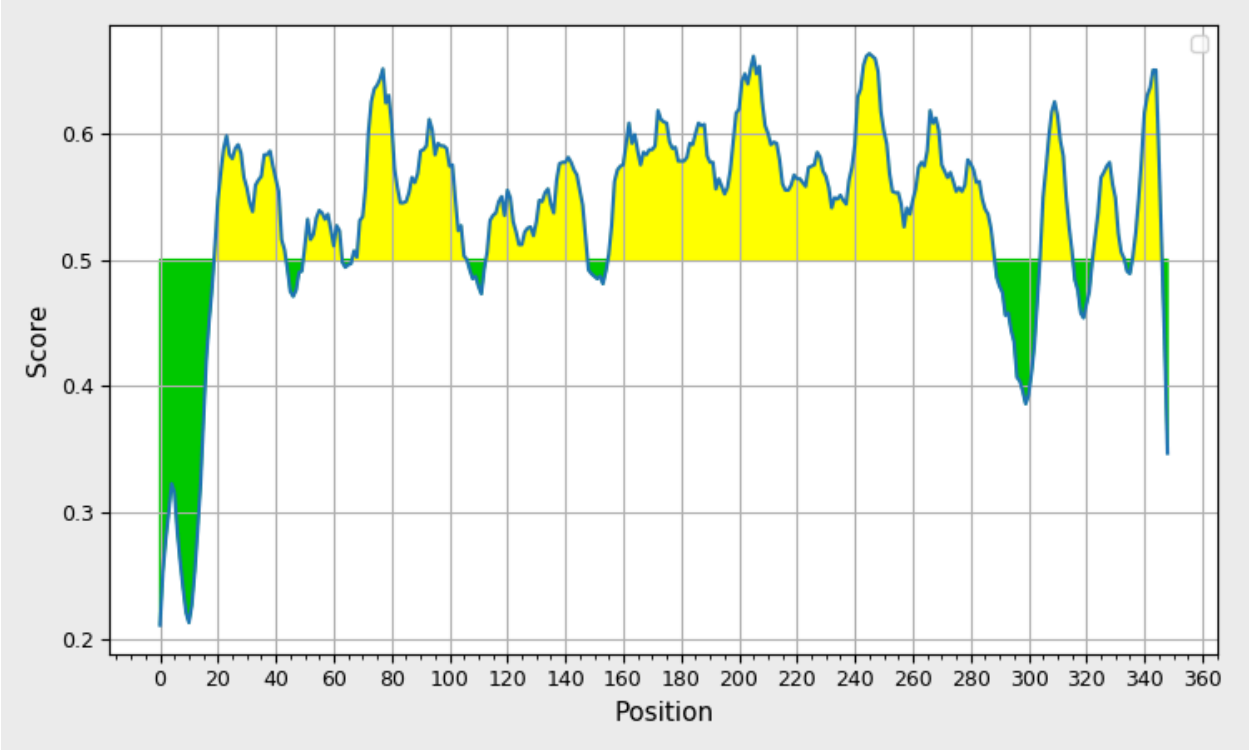
LEAIHSDKRISFPMYDDLIEQCHLSMEHKSKLVDKALNKLESTIDSQSRLSYLPPEIMRNIITKLSDYHLNSMLYGK

NHYKYYP

Predicted peptides & Scores:

No	Start	End	Peptide	Length
1	20	44	RDIAPHAPSNGKCKDNEYRSRNLCC	25

2	51	63	TYAFRLCDSKTNT	13
3	68	106	CGSDTFTSHNNHLQACLSCNGRCDSNQVETRSCNTTHNR	39
4	114	148	YYCLLKGASGCRTCISKTKCGIGYGVSGYTSTGDV	35
5	156	289	GTYSHTVSSTDKCEPVVTSNTFNYIDVEINLYPVNDTSCTRTTTTGLSESISTSELTITMNHKDCD PVFRAEYFSVLNNVATSGFFTGENRYQNTSKICTLNFEIKCNNKDSSSKQLTKTKNDTIMPHSETV TL	134
6	305	316	NTNTQDYENDTI	12
7	324	334	LDVNSHMPASC	11
8	337	347	HKLITNSQNPT	11



Protein : 4

Input Sequences : MPXV|URK20443.1|Ankyrin (CPXV-017) D1L - Probable NON-ANTIGEN without B-Cell Epitopes

>MESVDFMAVDEQFHDDLDLWSLSLVDDYKKHGLGVDCYVLEPVV

DRKIFDRFLLEPICDPVDVLYDYFRIHRDNIDQYIVDRLFAYITYKDIISALVSKNYM

EDIFSIIIKNCNSVQDLLLYYLSNAYVEIDIVDLMVDHGAVIYKIECLNAYFRGICKK

ESSVVEFILNCGIPDENDVKLDLYKIIQYTRGFLVDEPTVLEIYKLCIPYIEDINQLD

AGGRTLLYRAIYAGYIDLVSWLLENGANVNAVMSNGYTCLDVAVDRGSVIARREAHLK

ILEILLREPLSIDCIKLAILNNTIENHDVIKLCIKYFMMVDYSLCNVYASSLFDYIID

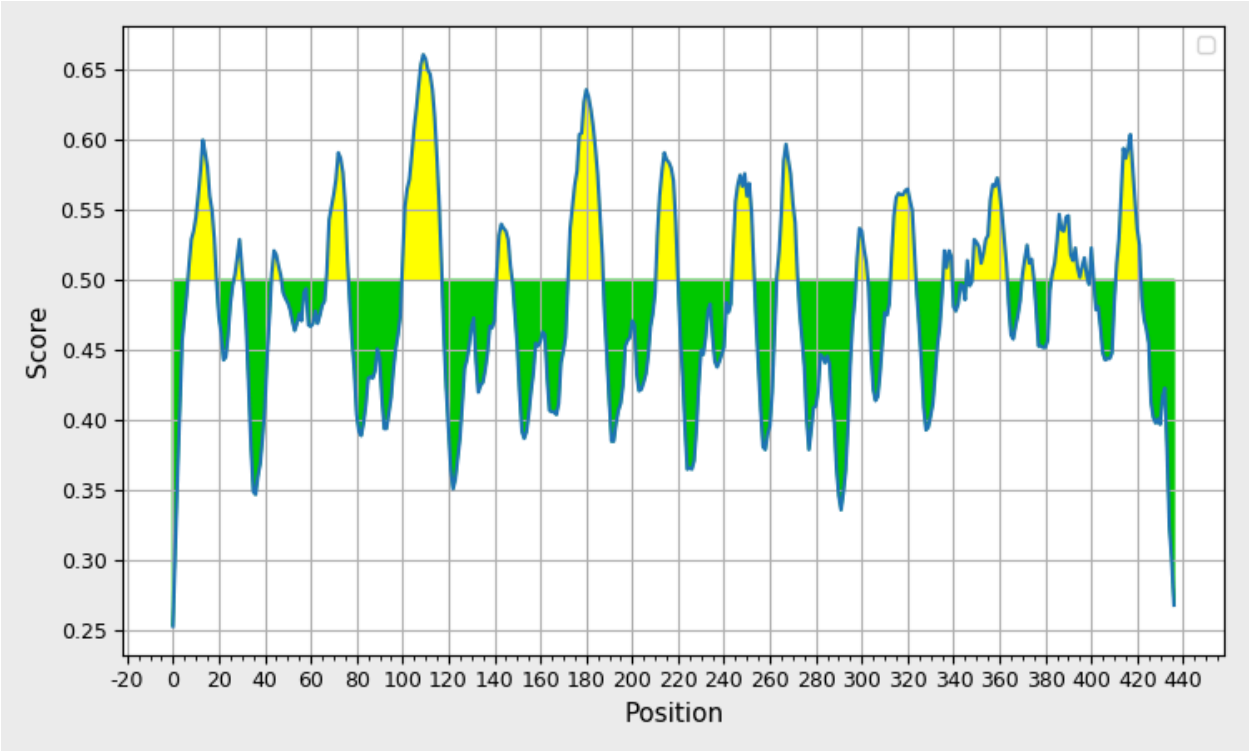
CKQELEYIRQMKIHNTTMYELIYNRDKNKHASHILHRYSKHPVLTQCITKGFKIYTEV

TEQVTKALNRRALIDEIINNVSTDDNLLSKLPLEIRDLIVSQAVI

Predicted peptides:

1	8	19	AVDEQFHDDLDL	12
2	28	31	YKKH	4
3	44	48	VDRKI	5
4	68	77	FRIHRDNIDQ	10
5	101	118	YMEDIFSIIIKNCNSVQD	18
6	142	148	AVIYKIE	7
7	174	188	PDENDVKLDLYKIIQ	15
8	212	220	EDINQLDAG	9
9	245	254	ANVNAVMSNG	10
10	264	273	RGSVIARREA	10
11	299	303	TIENH	5
12	314	324	MMVDYSLCNVY	11
13	337	340	QELE	4
14	347	347	I	1

15	350	364	TTMYELIYNRDKNKH	15
16	371	376	RYSKHP	6
17	384	399	KGFKIYTEVTEQVTKA	16
18	401	402	NR	2
19	412	422	NVSTDDNLLSK	11



Protein : 5

Input Sequences : MPXV|URK20444.1|Host range protein; D2L

>MYYANICLDFDNNVYTVKDKNYTNAVIEYPVVCNFRRYSESESD

SDIDDRAELPNAIMIIFIMVLENMIKNY

Predicted peptides:

1	15	23	YTVKDKNYT	9

2	36	52	RRYSESESDSDIDDRAE	17

Protein : 6

Input Sequences : MPXV|URK20445.1|D3R Secreted EGF-like protein

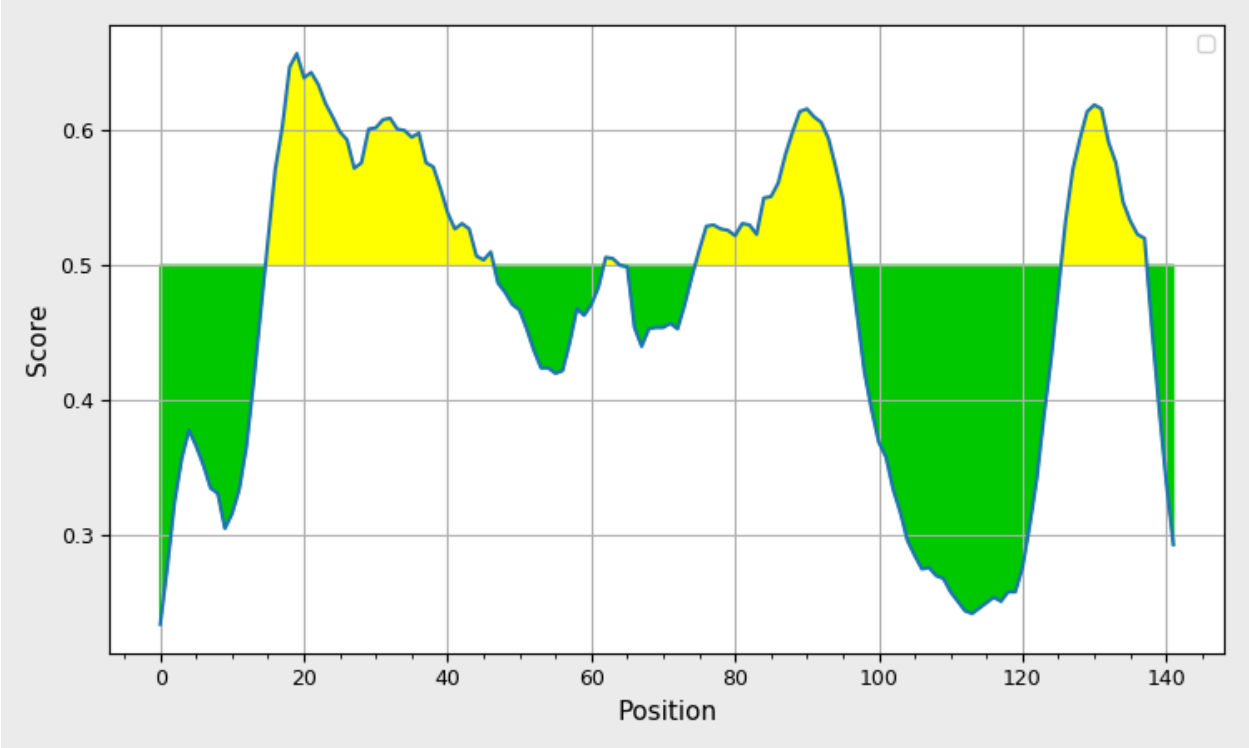
>MSIKYLMLLFAAMIIRSLADSGNAIETTSPEITNATTDIPAIRL

CGPEGDGYCLHGDCIHARDIDGMYCRCSHGYTGIRCQHVVLVDYQRSGKPDTTTSYIP

SLGIVLVLVGIIITCCLLSVYMFTRRTKLPIQDMVVLYFL

Predicted peptides:

1	16	47	RSLADSGNAIETTSPEITNATTDIPAIRLCGP	32
2	63	65	DID	3
3	76	97	TGIRCQHVVLVDYQRSGKPDTT	22
4	127	138	RRTKLPIQDMVV	12



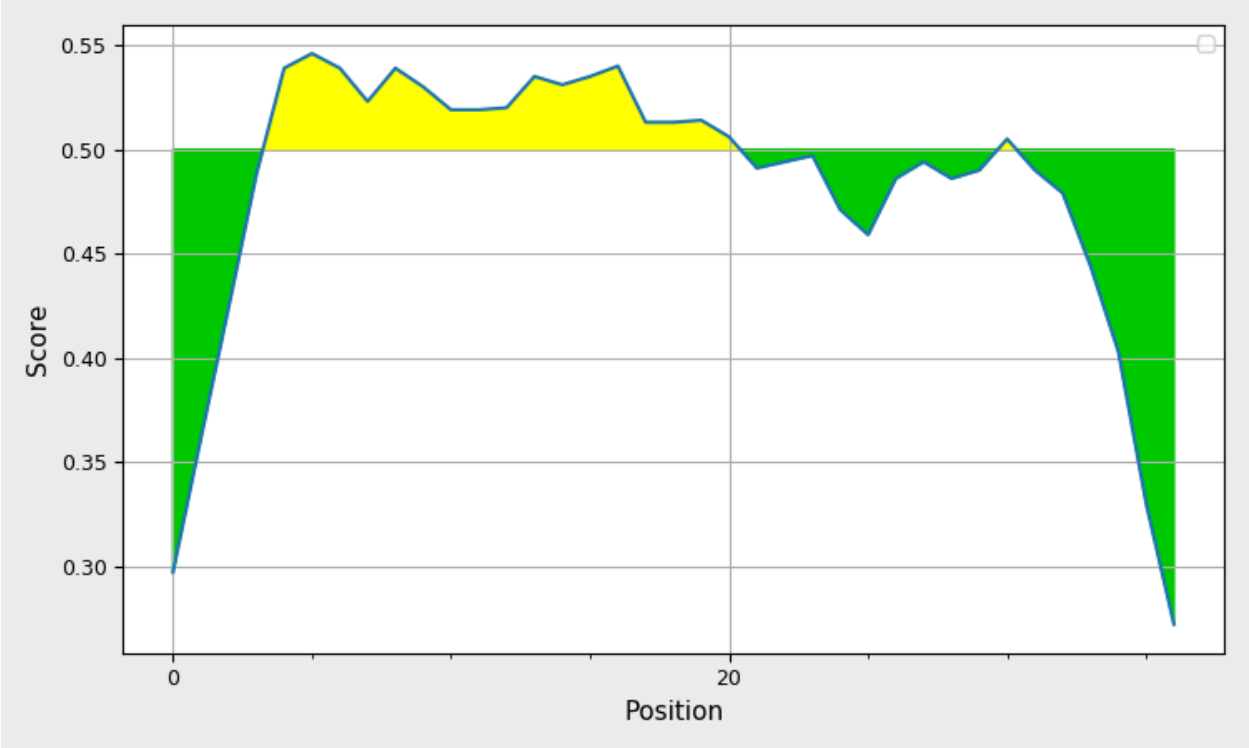
Protein : 7

Input Sequences : MPXV|URK20446.1|IL-1 receptor antagonist

>MDMYDDKGLQTIKLFNNEFDCIRNDIRELFKHVTDSE

Predicted peptides:

1	5	21	DDKGLQTIKLFNNEFDC	17
2	31	31	K	1



Protein : 8

Input Sequence : MPXV|URK20447.1|D5R Zinc finger-like protein inhibition of apoptosis induced by UV irradiation virosome localization virulence factor zinc binding

>MEFDPVKINTSSIDHVTILQYIDEPNDIRLTVCIIRNVNNITYY

INITKINPHLANRFRAWKKRIAGRDYMTNLSRDTGIQQSKLTETIRNCQKNKNIYGLY

IHYNLVINVVIDWITDVIVQSILRGLVNWYIANNTYTPNTPNNTTTISELDIIKILDK

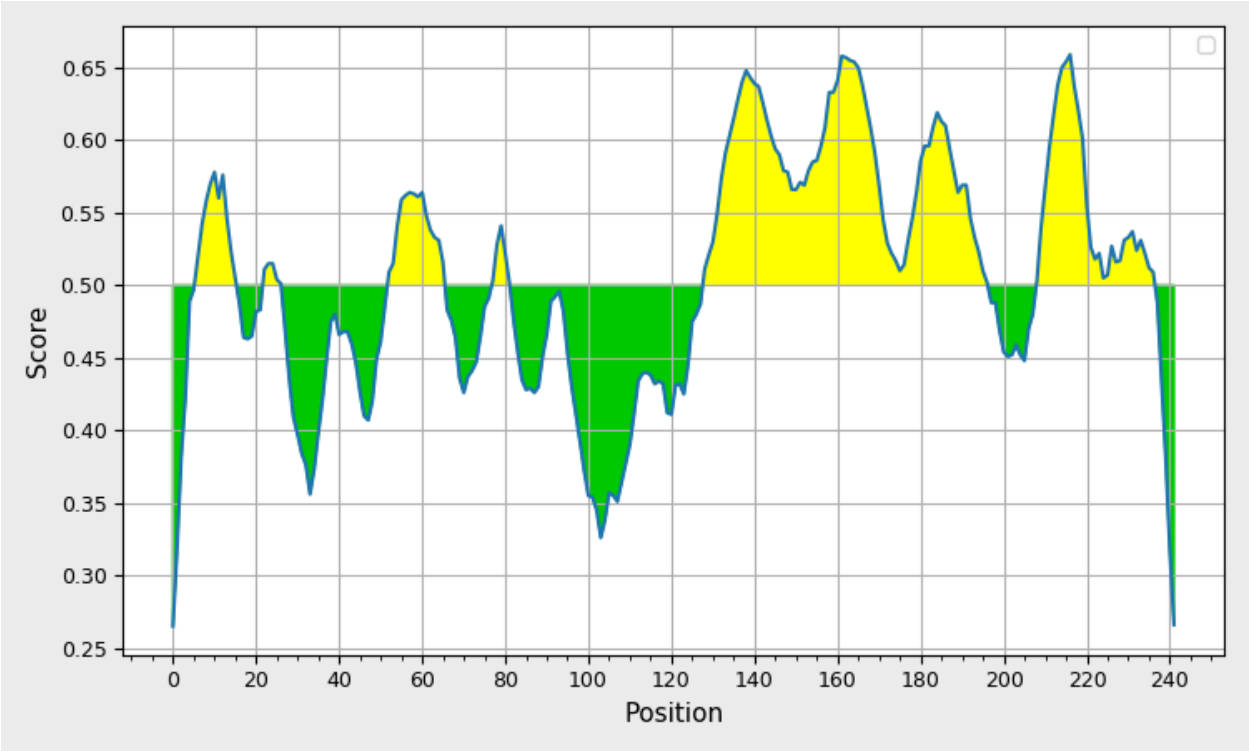
YEDMYRVSKEKECGICYEVVYSKRLENDRYFGLLDSCNHIFCITCINIWHRTRRETGA

SDNCPICRTRFKKITMSKFYKLVN

Predicted peptides:

1	7	16	KINTSSIDHV	10
2	23	27	DEPND	5
3	53	66	HLANRFRAWKKRIA	14
4	78	82	TGIQQ	5
5	129	197	VNWYIANNTYTPNTPNNTTTISELDIIKILDKYEDMYRVSKEKECGICYEVVYSKRLEN DRYFGLLDSC	69

6	209	237	WHRTRRETGASDNCPICRTRFKKITMSKF	29



Protein : 9

Input Sequences : MPXV|URK20448.1|D6L Soluble IL-18 binding protein (Bsh-D7L)

secreted IL-18 binding protein

>MRILFLIAFMYGCVHSYVNAVETKCPNLAIVTSSGEFHCSGCVE

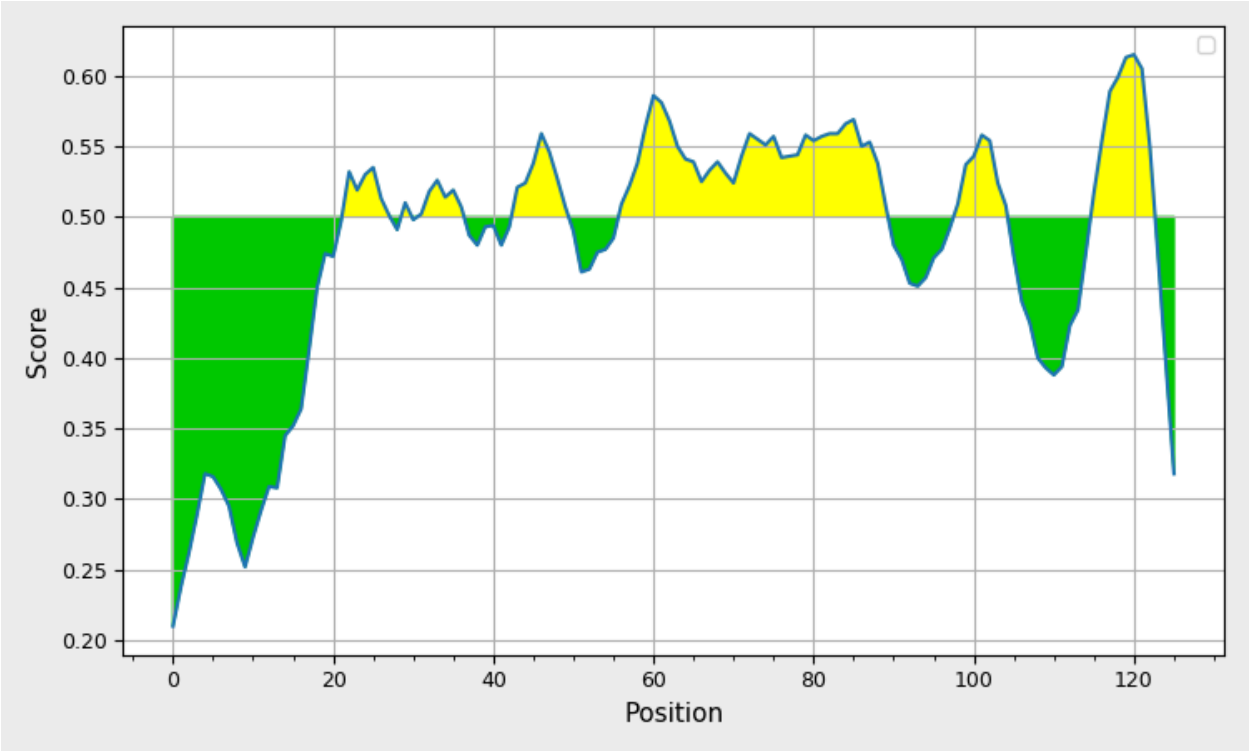
RMPGFSYMYWLANDMKSDEDTKFIEHLGGGIKEDETVRTTDGGITTLRKVLHVTDTNK

FAHYRFTCVLITLDGVSKKNIWLK

Predicted peptides:

1	23	28	TKCPNL	6
2	30	30	I	1
3	32	37	TSSGEF	6
4	44	50	ERMPGFS	7
5	57	90	NDMKSDEDTKFIEHLGGGIKEDETVRTTDGGITT	34
6	99	105	DTNKFAH	7

7	116	123	DGVSKKNI	8



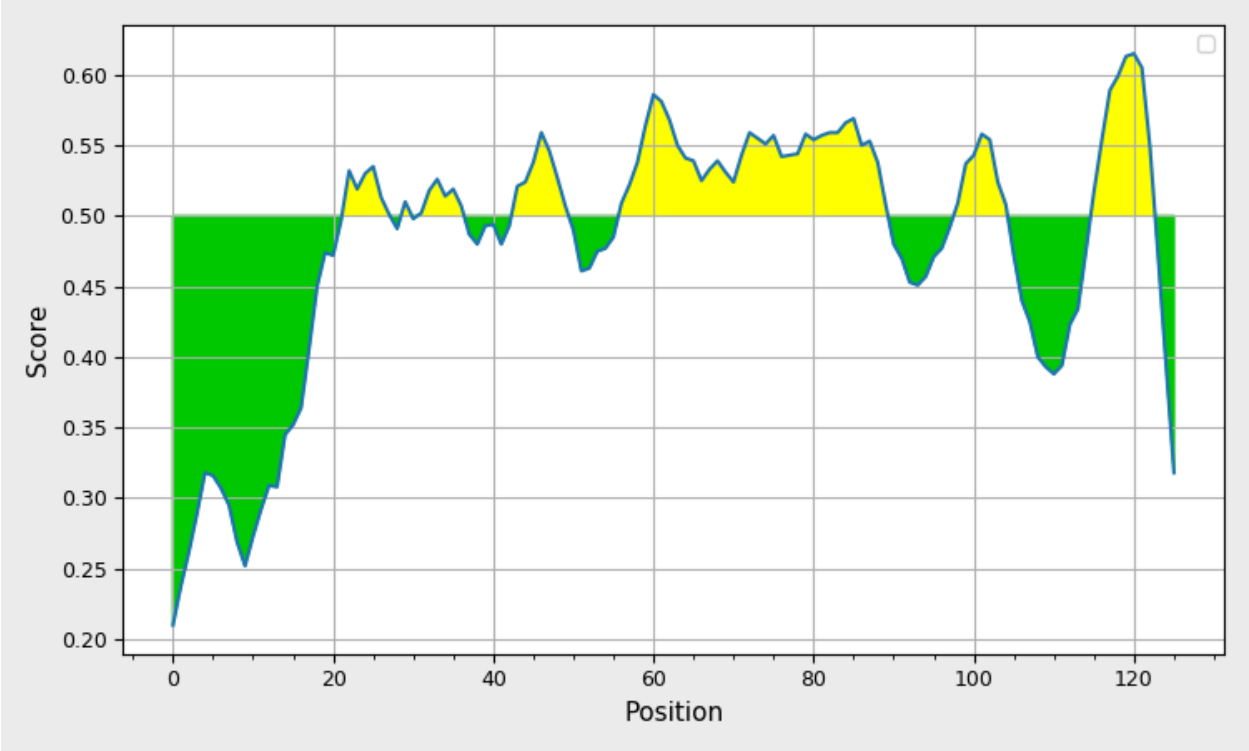
Protein : 10

Input Sequences : MPXV|URK20449.1|Ankyrin/Host Range (Bang-D8L) D7L ankyrin-like host range

>MFDYLENEEVVLDELKQMLRDRDPNDTRNQFKNNALHAYLFNEH
CNNVEVVKLLLDSGTNPLHKNWRQFTPIEEYTNSRHVKVNKDIAMALLEATGYSNIND
FNIFSYMKSKNVDVDLIKVLVEHGFDLSVKCENHRSVIENYVMTDDPVPEIIDLFIEN
GCSVLYEDDEYGYVYDDYQPRNCGTVLHLYIIAHLYSESDTRAYVRPEVVKCLINHGI
KPSSIDKNYCTALQYYIKSSHIDIDIVKLLMKGIDNTAYSYIDDLTCCTRGIMADYLN
SDYRYNKDVDLDLVKLFLENGKPYGIMCSIVPLWRNDKETISLILKTMNSDVLQHILI
EYMTFGDIDIPLVECMLEYGAVVNKEAIHRYFRNINIDSYTMKYLLKKEGGDAVNHLD
DGEIPIGHLCESNYGCYNFYTYTYKKGLCDMSYVCPILSTINICLPYLKDINMIDKRG
ETLLHKAVRYNKQSLVSLLLESGSDVNIRSNNGYTCITIAINESRNIELLKMLLCHKP
TLDCVIDSLSEVSNIVDNAYAIKQCIKYTMIIDDCTSSKIPESISQRYNDYIDLCNQE
LNEMKKIMVGGNTMFSLIFTEHGAKIIHRYANNPELRAYYESKQNKIYVEAYDIISDA
IVKHNKIHKTIIKSVDDNTYISNLPYTIKYKIFEQQ

Predicted peptides:

1	23	28	TKCPNL	6
2	30	30	I	1
3	32	37	TSSGEF	6
4	44	50	ERMPGFS	7
5	57	90	NDMKSDEDTKFIEHLGGGIKEDETVRTTDGG ITT	34
6	99	105	DTNKFAH	7
7	116	123	DGVSKKNI	8



Protein : 11

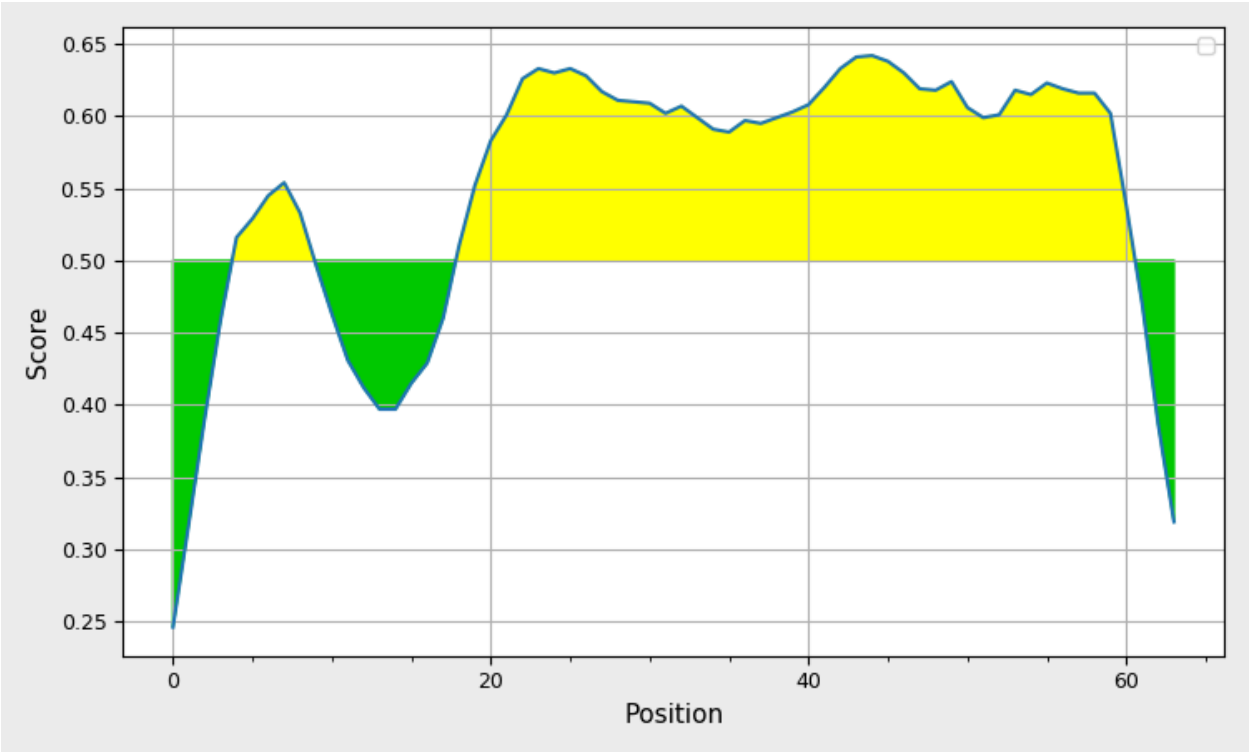
Input Sequences : MPXV|URK20450.1|ANK-containing protein D8L

>MSSKGGSSGGMWSVFIHGHDGSNKGSKTYTSSGSGMWGGGSSSG

VKSGVNGGVNGGVKSGTGKI

Predicted peptides:

1	5	9	GGSSG	5
2	19	61	HDGSNKGSKTYTSSGSGMWGGGSSSGVKSGVNGGVNGGV KSGT	43



Protein : 12

Input Sequences : MPXV|URK20451.1|Ankyrin D9L Type I IFN resistance

>MVNDKILYDSCKTFNIDASSAQSLIESGANPLYEYDGETPLKAY

VTKKNNNIKNDVVILLLSSVDYKNINDFDIFEYLCSDNIDIDLLKLLISKGIEINSIK

NGINIVEKYATTSNPNIDVFKLLLDKGIPTCSNIQYGYNIIPIQCTDYIYFNWDDEFD

YLDYDYTTDYDNRMGKTVLYYYIITRSQDGYVTSLDVINYLISHENEMCHYTYRERTI

LYYYVDKCDIKREIFDVLFDSNYSGNELMHILSIYLRKQYRKKNHKIDNYIVDKLLSA

HDTFYILELCNSLRNNVIISSILKRYTDSIQDLLSEYVSYHTVYINVIKCMIREGAIL

YRFKHINKYFKRFDNRDPKVVEYILKNGNVVVNDDNIINIMPLFPTLFIHESEVLSIL

EICKPYIDDINKIDKHGRSILYYCIESHSVDLIEWLIDNGADINITTTYGSTCIGICV

IMAHACIPEIAEIYIKILEIILSKLPTIECIKKTVDYLSNDRHLLIGNKAKSLLKICI

KYFILVDYKYICDTYPSYIEYITDCEKEIADMCQIKINGTDMLTVMYKLNKPTKKRYV

NNPIFTDWANKQYKFYNQIIYNANRLIEQSKKIDNMIDEVSADNNRLSTLPLELRHLI

FSYAFL

Predicted peptides:

1	15	16	NI	2
2	18	18	A	1
3	29	37	ANPLYEYDG	9
4	49	51	NNN	3
5	64	72	VDYKNINDF	9
6	96	105	IEINSIKNGI	10
7	131	146	PTCSNIQYGYNIIPIQ	16
8	159	160	FD	2
9	167	178	TTDYDNRMGKTV	12
10	190	195	GYVTSL	6
11	206	214	NEMCHYTYR	9
12	234	234	D	1
13	236	245	LFDSNYSGNE	10
14	259	260	RK	2
15	268	278	YIVDKLLSAHD	11
16	303	312	TDSIQDLLSE	10
17	332	341	AILYRFKHIN	10
18	360	372	KNGNVVVNDDNII	13
19	386	386	S	1
20	400	408	DDINKIDKH	9
21	433	441	ADINITTTY	9
22	455	463	ACIPEIAEI	9
23	477	489	TIECIKKTVDYLS	13

24	512	521	ILVDYKYICD	10
25	538	538	A	1
26	540	545	MCQIKI	6
27	559	581	KPTKKRYVNNPIFTDWANKQY KF	23
28	583	583	N	1
29	587	598	YNANRLIEQSKK	12
30	601	601	N	1
31	604	614	DEVSADNNRLS	11

Protein :13

Input Sequences : MPXV|URK20452.1|D10L Type 1 IFN inhibitor

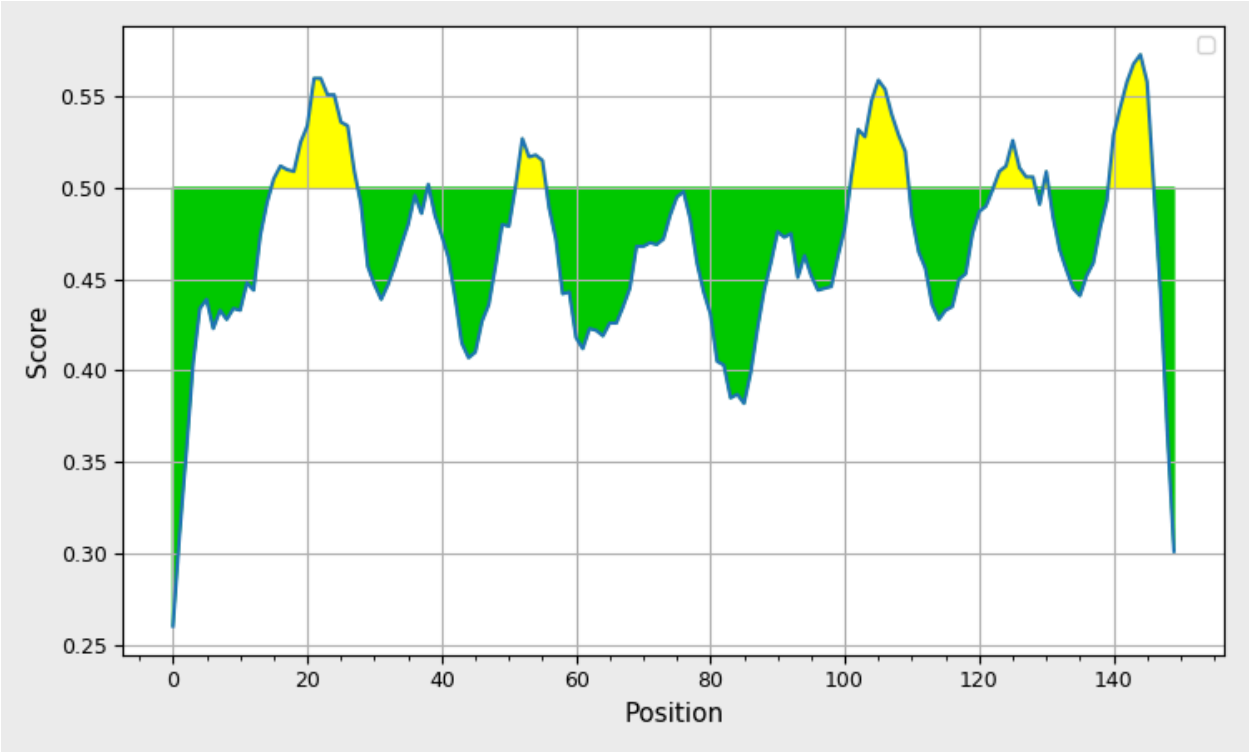
>MGIQHEFDIIINGDIALRNLQLHRGDNYGCKLKIISNDYKKLKL

RFIIRPDWSEIDEVKGLTVFANNYAVKVNKVDYTFYYVIYEAVIHLYNKKTEILIYSD

DENELFKHYYPYISLNMISKKYKVKEENYSSPYIEHPLIPYRDYESMD

Predicted peptides:

1	16	28	ALRNLQLHRGDNY	13
2	39	39	Y	1
3	52	56	WSEID	5
4	102	110	DDENELFKH	9
5	124	129	YKVKEE	6
6	131	131	Y	1
7	141	146	IPYRDY	6



Protein : 14

Input Sequences : MPXV|URK20453.1|Bcl-2-like protein, IFN-beta inhibitor

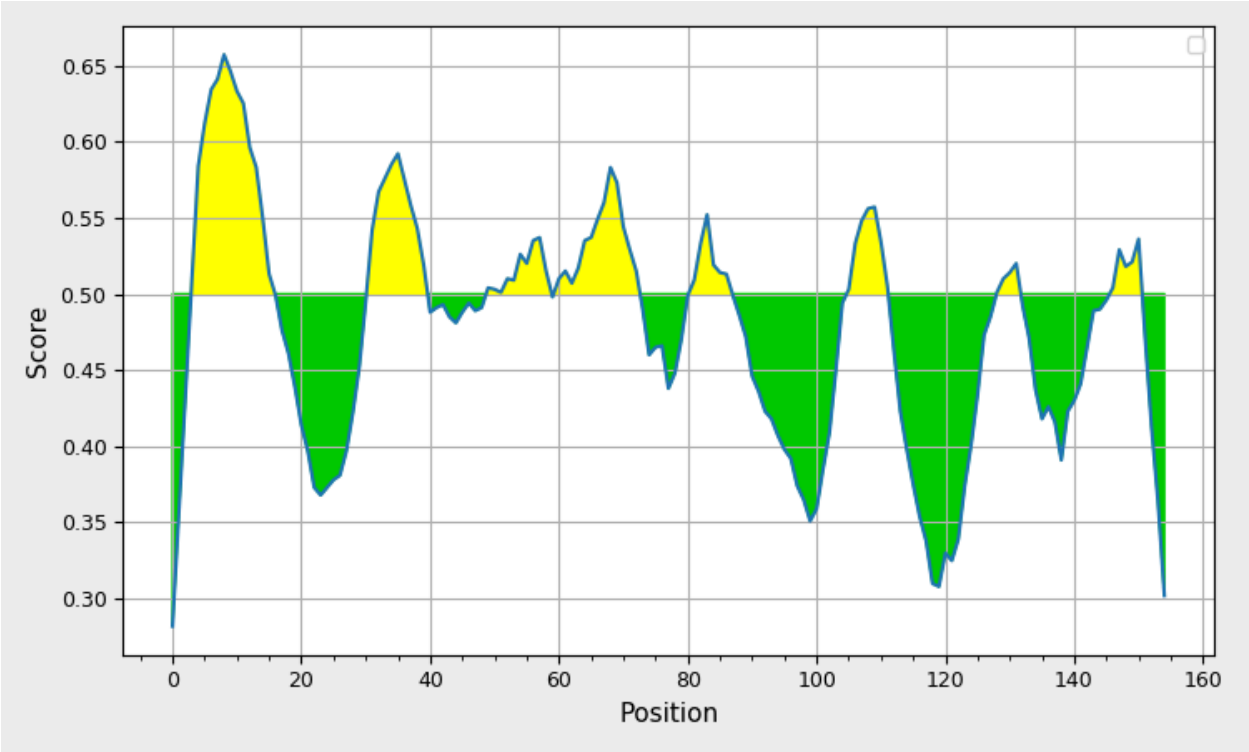
>MNVYNKADSFSLESDSIKDVIHDYICWLSMTDETRPSIGNVFTA

METFKIDAVRYYDGNIYDLAKDINTMSFDSFIRSLQNITSKKDKLTVYGTMGLLSIVV

DINKGRDVSNIKFAAGIIILMEYIFDDTDLSHLKIALYRRIQRCYPIDDDDDR

Predicted peptides:

1	4	16	YNKADSFSLESDS	13
2	32	40	DETRPSIGN	9
3	50	59	IDAVRYYDGN	10
4	61	73	YDLAKDINTMSFD	13
5	82	87	ITSKKD	6
6	106	112	KGRDVSN	7
7	129	132	DTDL	4
8	147	151	YPIDD	5



Protein : 15

Input Sequences : MPXV|URK20454.1|BTB domain of kelch-like protein D12L Kelch-like protein

>MTSLIMDMNRLNLDKLKHENIFSDNIIEDAKEFIFGSRKIYTDS

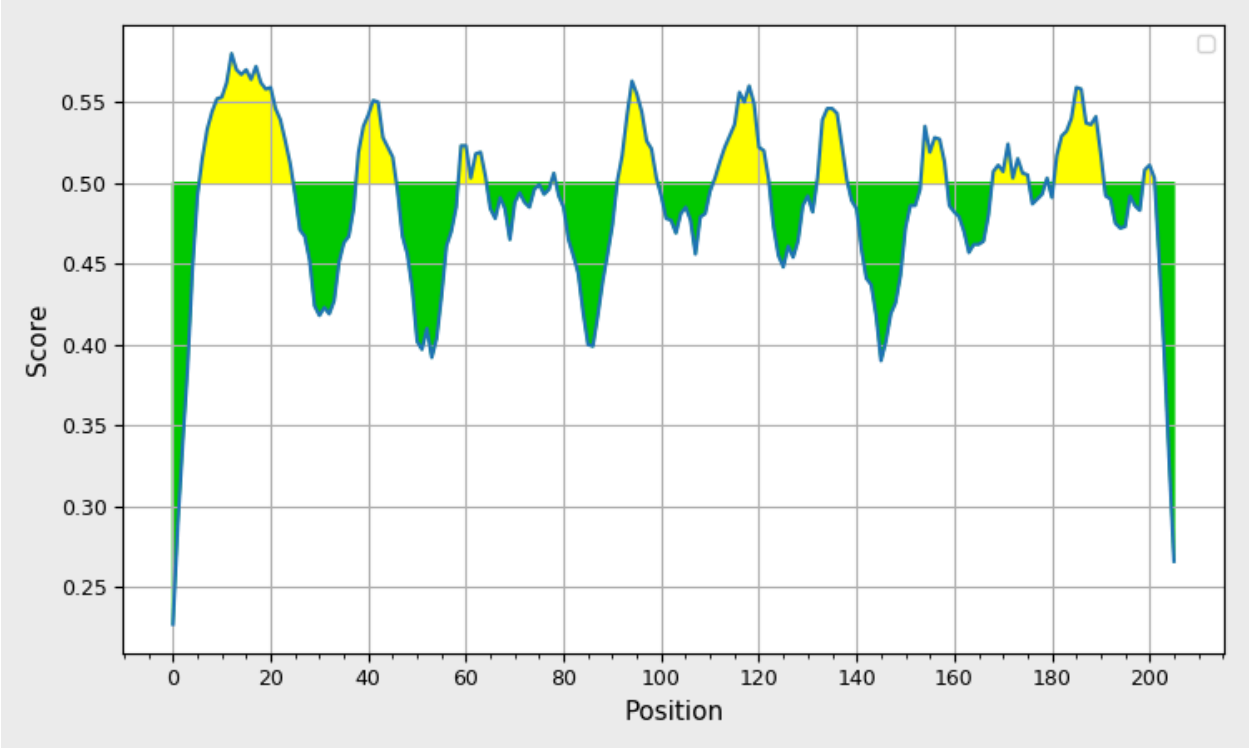
VDDLIELYSLAKYLNNQTLKDVVIERMDYVCKYIGKDNWSTIYSFYKENGLRNSFLRQ

YINNNIEEICSTDQFLKLDVDSVCDILDNDEIVVTREYTILNMVLRWLENKRVNIDDF

TKVMFVIRFKFITYSELTNAIEKIAPEYRQRLQDLYHKKLRVLDIL

Predicted peptides:

1	7	25	DMNRLNLDKLKHENIFSDN	19
2	39	46	KIYTDSVD	8
3	60	65	NQTLKD	6
4	79	79	G	1
5	92	100	ENGLRNSFL	9
6	112	123	CSTDQFLKLDVD	12
7	133	139	EIVVTRE	7
8	155	159	VNIDD	5
9	169	176	FKFITYSE	8
10	180	180	A	1
11	182	191	EKIAPEYRQR	10
12	200	202	LRV	3



Protein : 16

Input Sequences : MPXV|URK20455.1| D13L IL-1 receptor antagonist

>MDTIKIFNHGEFDTIRNELVNLLKVVKWNTINSNVTVSSTDTID

ISDCIREILYKQFKNVRNIEVSSDISFIKYNRFNDTTITDDNMGYYLVIYLNRTKSLK

TLIYPTLETVISSSDDIMFSKSLNFRFENTKREYKLVMCSISLIYKPSICRIQYDNNK

YIDISDSQEGNNLCYCVITMDPHHLIDLETMCVLVDKSGKCLLVNEFYIRFRKNHIYN

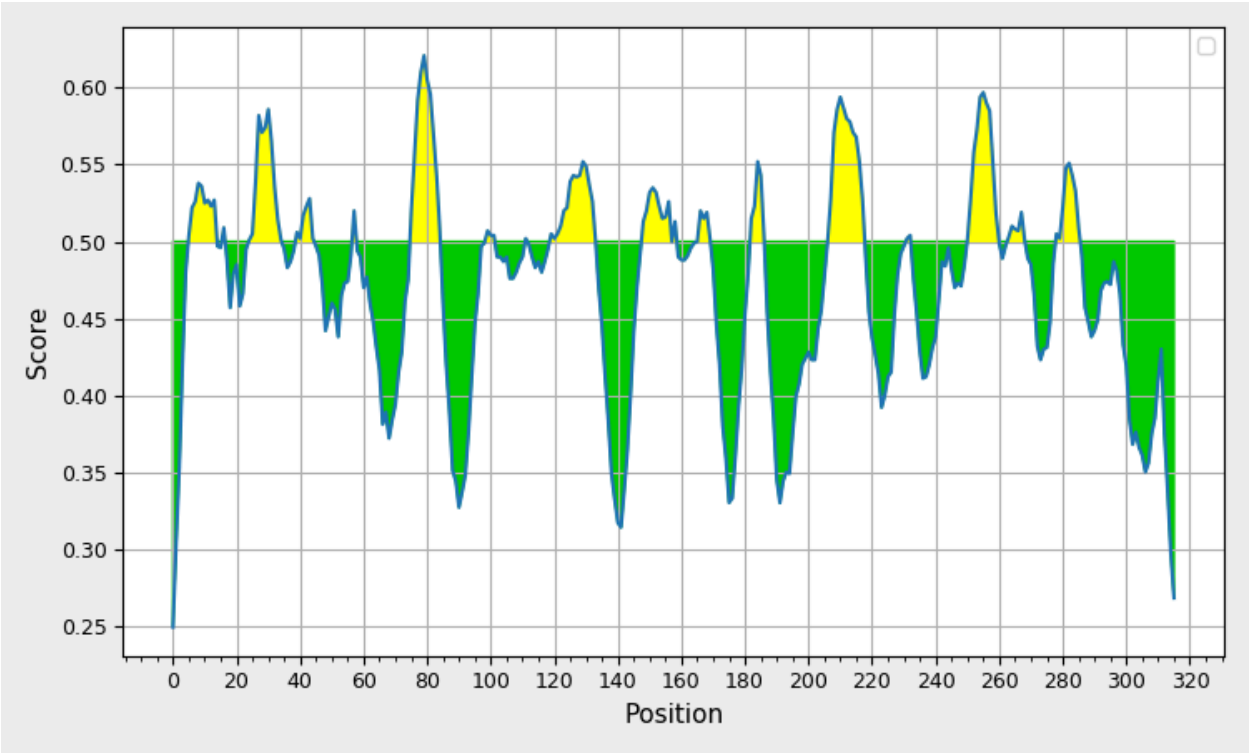
SFVDICMDHIFELPNTKELFTLCNDDGRNIAWDNDKLESGNNTWIPKTDDEYMFLSKL

MNIAKFNNTKFDYYVLVGDTDPCTVFTFKVTKYYININYE

Predicted peptides:

1	6	14	IFNHGEFDT	9
2	17	17	N	1
3	25	35	VVKWNTINSNV	11
4	40	45	TDTIDI	6
5	58	58	K	1
6	76	85	RFNDTTITDD	10

7	100	102	SLK	3
8	112	112	V	1
9	120	133	MFSKSLNFRFENTK	14
10	149	157	PSICRIQYD	9
11	159	159	N	1
12	166	170	DSQEG	5
13	183	187	HHLID	5
14	208	218	YIRFRKNHIYN	11
15	232	233	PN	2
16	251	260	DNDKLESGNN	10
17	264	269	PKTDDE	6
18	279	286	IAKFNNTK	8



Protein : 17

Input Sequences : MPXV|URK20456.1|D14L IL-1 receptor antagonist

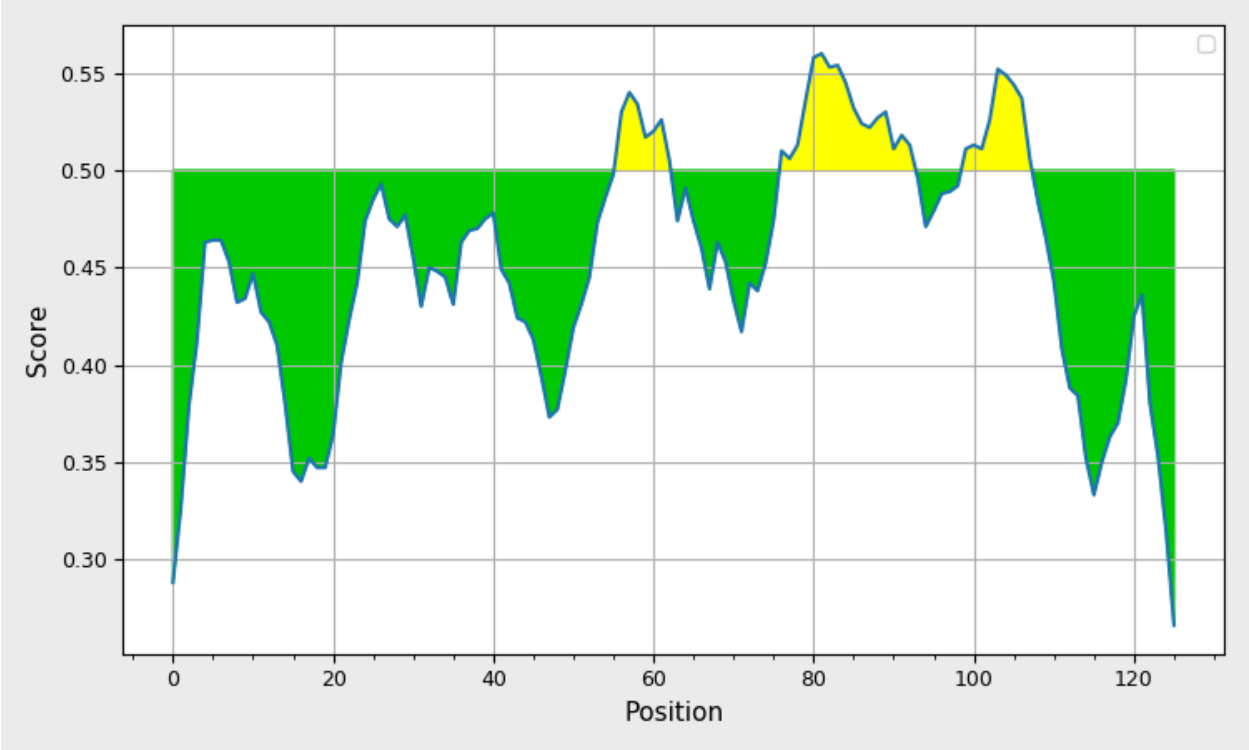
>MRWKKINITINNVEMILVAAIIIDVPPVIDLCVKTMIHNINSTN

CIRMFNFSKRYGIKKLYNTSMSEIINNITAVTSDPGTYFQNVYTNKHKKYTMVYVRSD

IKWILLKYLIMESLILLEMNWSTC

Predicted peptides:

1	57	63	IKKLYNT	7
2	77	93	SDPGTYFQNVYTNKHKK	17
3	100	108	RSDIKWILL	9



Protein : 18

Input Sequencs : MPXV|URK20457.1|D19L Putative TLR signalling inhibitor

>MIMRTDTNNILMRHLKNLTDDEFKCIIHRSSDFLYLSDRDYTSI

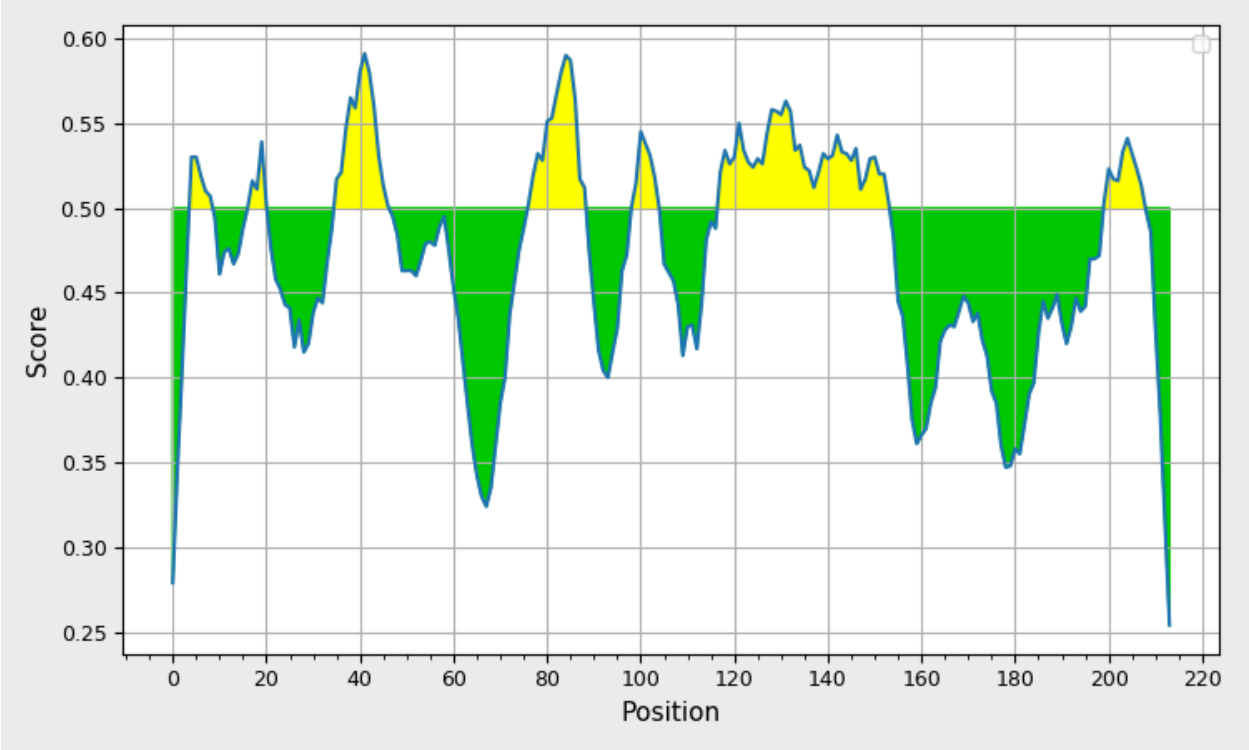
TKETLVSEIVEEYPDDCNKILAIIFLVLDKDIDVDIETKLTPTPKPAVRFAILDKMTA

DIKLTDLVRHYFRYIEQDIPLGPLFKKIDSYRIRAINNYSKELGLATEYFNKYGHLMF

YTLPIPYNRFFCRNSIGFLAVLSPTIGYVKAFYKFIEYVSIDDRLKFKKELMSK

Predicted peptides:

1	5	9	TDTNN	5
2	17	21	NLTDD	5
3	36	47	LSDRDYTSITKE	12
4	77	89	DVDIETKLTPTPK	13
5	99	104	KMTADI	6
6	118	154	EQDIPLGPLFKKIDSYRIRAINNYSKELGLATEY FNK	37
7	200	208	SIDDRLKFK	9



Protein : 19

Input Sequences : MPXV|URK20458.1|Anti-apoptotic Bcl-2-like protein

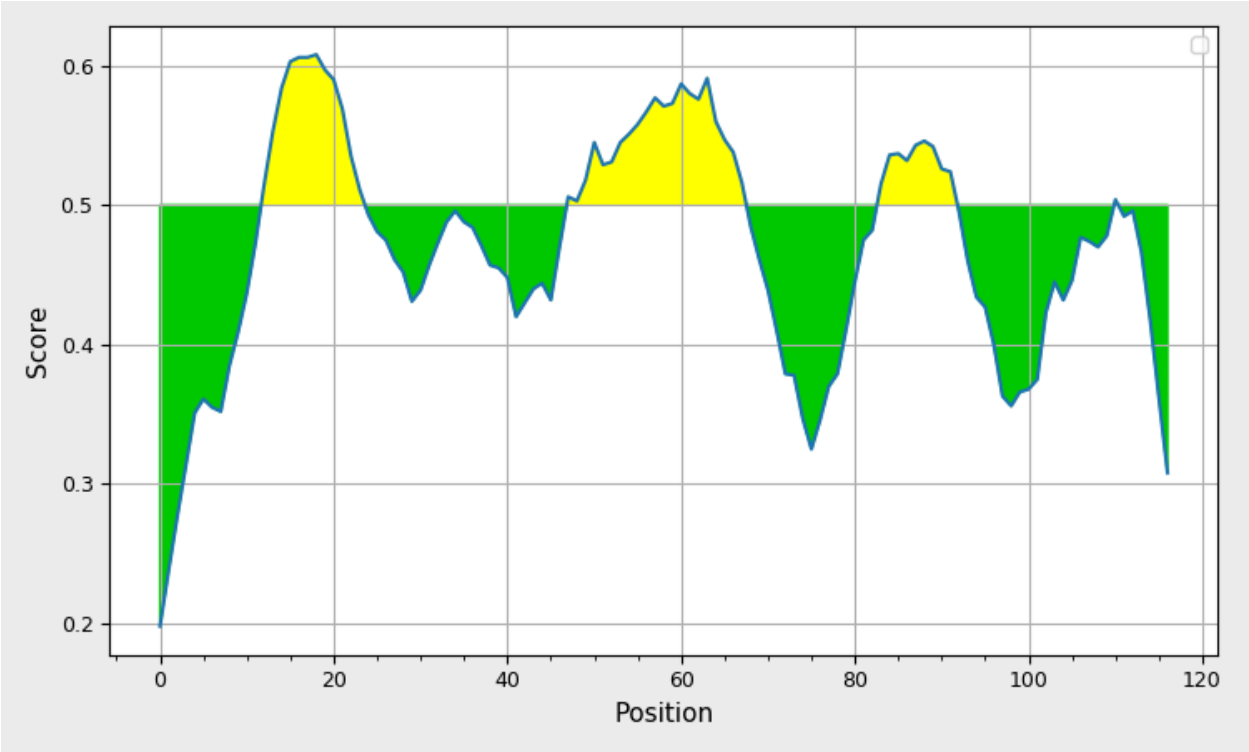
>MRTLLIRYILWRNDGDPSYYNDDFKKLILFDELVDDDDVCTLIK

NMRMTLSDGPLLDRLNQPVNNVEDVKRMIAISAKVARDIGRRSEIRWEDSFTILFRMI

EKYFDDLMTDLYGEK

Predicted peptides:

1	13	24	NDGDPSYYNDDF	12
2	48	68	MTLSDGPLLDRLNQPVNNVED	21
3	84	92	GRRSEIRWE	9
4	111	111	T	1



Protein : 20

Input Sequences : MPXV|URK20459.1|Alpha amanatin target protein

>MTSSAMDNNESKVLEMVYDATILPDCSGMDPSIIDCINRHINMR

IQRSYSSNIIAILDRFLMMNKDELNNTQCHIIKEFMTYEQMAIDYYGGYVNAILYQIR

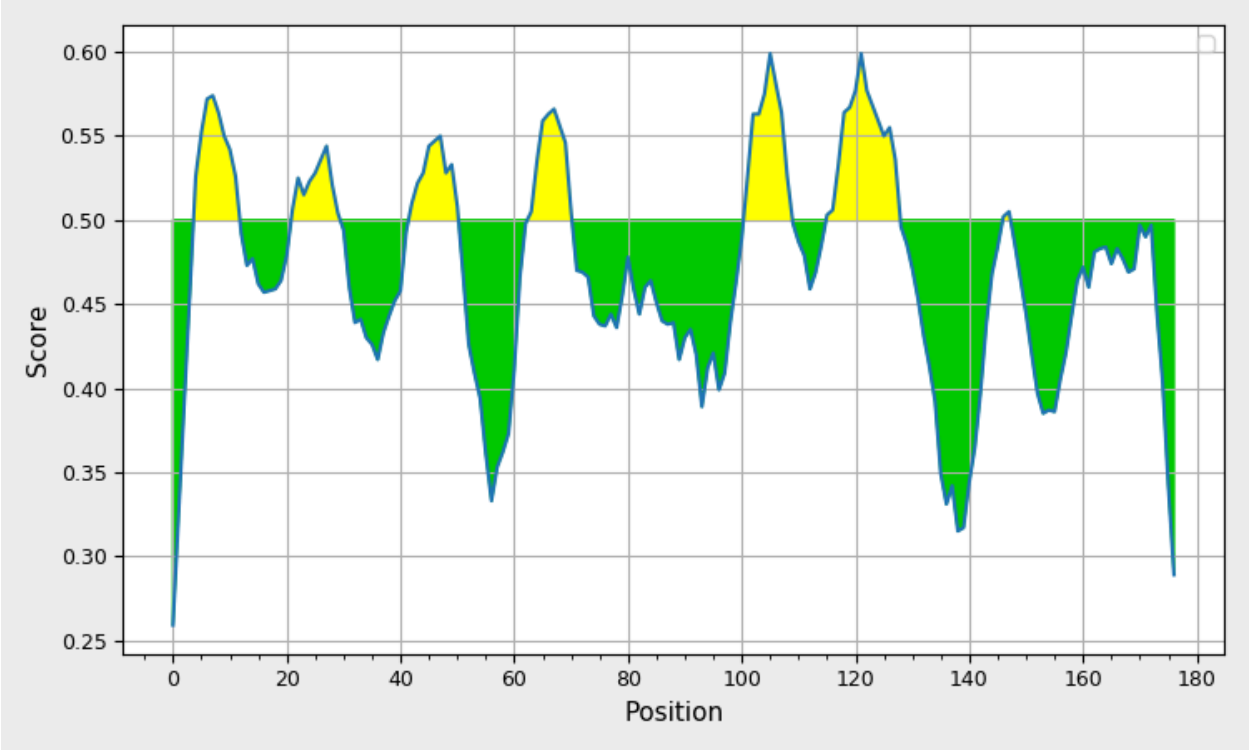
KRPNQHHTIDLFKRIKRTRYDTFKVNPVEFVKKVIGFVSILNKYKPIYSYVLYENVLY

DELKCFIDYVETKYFQN

Predicted peptides:

1	5	12	AMDNNESK	8
2	22	30	ILPDCSGMD	9

3	43	51	MRIQRSYSS	9
4	64	71	NKDELNNT	8
5	102	109	RKRPNQHH	8
6	116	128	RIKRTRYDTFKVN	13
7	147	148	KP	2



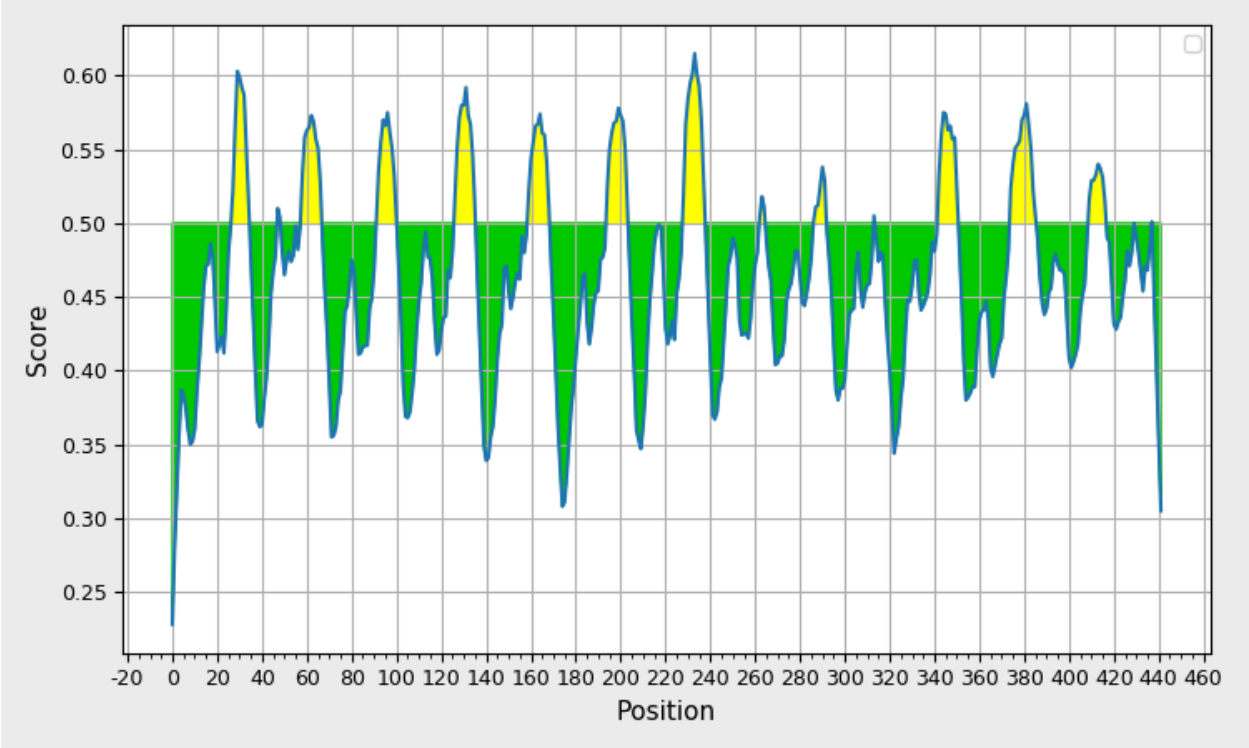
Protein : 21

Input Sequences : MPXV|URK20460.1|ANK-containing protein O1L ankyrin-like apoptosis inihibitor

>MSVEYYLSLYAKYNSKNLDVFRNMLQVIEPSGNYHILHAYCGIK
GLDERFIEELLHRGYSPNETDDDGNYPLHIASKINNNRIVAMLLVHGADPNACDKHNK
TPLYYLSGTDDEVIERINLLVQYGAKINNSVDEEGCGPLLACTDPSERVFKKIMSIGF
EARIVDKFGKNHIHRHLMSDNPKASTISWMMKLGISPSKPDHDGNTPLHIVCSKTVKY
VDIINLLLPSTDVNKQNKFGDSPLTLLIKTLSPAHLINKLISTSNVITDQTVNICIFY
DRDDVLEIINDKGKQYDSTDFKMAVEVGSIRCVKYLLDNDIICEDAMYYAVLSEYETM
VDYLLFNHFSVDSIVNGHTCMSECVRLNNPVILSKLMLHNPTSETMYITMKEIEKDKL
DKSIIIPFIAYFVLMHPDFCKNCRYFTSYKRFVTDYVHEGVSYEVFDDYF

Predicted peptides:

1	28	35	IEPSGNYH	8
2	48	49	ER	2
3	59	67	YSPNETDDD	9
4	92	101	ADPNACDKHN	10
5	127	136	AKINNSVDEE	10
6	160	169	FEARIVDKFG	10
7	195	204	ISPSKPDHDG	10
8	229	238	TDVNKQNKFG	10
9	263	265	NVI	3
10	287	292	DKGKQY	6
11	314	314	D	1
12	343	351	FSVDSIVNG	9
13	375	386	PTSETMYITMKE	12
14	410	417	DFCKNCRY	8
15	430	430	H	1
16	438	438	F	1



Protein : 22

Input Sequences : MPXV|URK20461.1|NFkB inhibitor

>MVYKLVLLFCIASLGYSVEYKNTICPHRQDYRYWYFVAELTIGV

NYDINSTIIGECHMSESYIDRNANIVLTGYGLKVNMTIMDTDQRFVAAAEGVGKDNKL

SVLLFTTQRLDKVHHNISVTITCMEMNCGTTKYNSDLPESIHKSSSCDITINGSCVTC

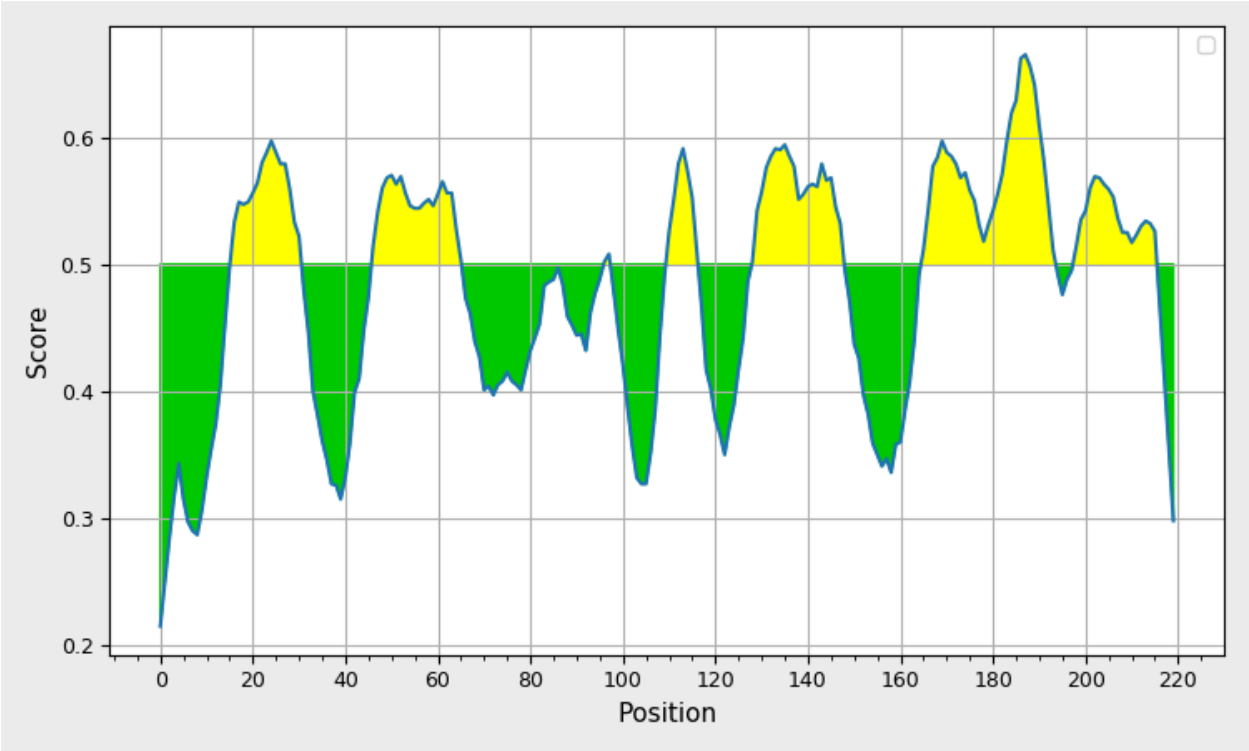
VNLETDPTKINPHYLHPKNKYLYHNSEYSMRGSYGVTFIDELNQCLLDIKELSYDICY

RE

Predicted peptides:

1	17	31	SVEYKNTICPHRQDY	15
2	47	66	DINSTIIGECHMSESYIDRN	20
3	97	98	GK	2
4	111	117	RLDKVHH	7
5	129	148	NCGTTKYNSDLPESIHKSSS	20
6	166	194	DPTKINPHYLHPKNKYLYHNSEYSMRGSY	29

7	199	216	IDELNQCLLDIKELSYDI	18



Protein : 23

Input Sequences : MPXV|URK20462.1|Ankyrin/NFkB inhibitor

>MDLSRINTWKSKQLKSFLSSKDAFKADINGHSVLYYAIADNNVR

LVCTLLNAGALKNLLDNEFPLHQAATLEDTKIVKILLFSGMDDSQFDDKGNTALYYAV

DSGNMQTVKLFVKKNWRLMFYGKTGWKTSFYHAVMLNDVSIVSYFLSEIPSPFDLAIL

LSCIHTTIKNGHVDMMILLLDYMTSTNTNSLLFIPDIKLAIDNKDIEMLQALFKYDIN

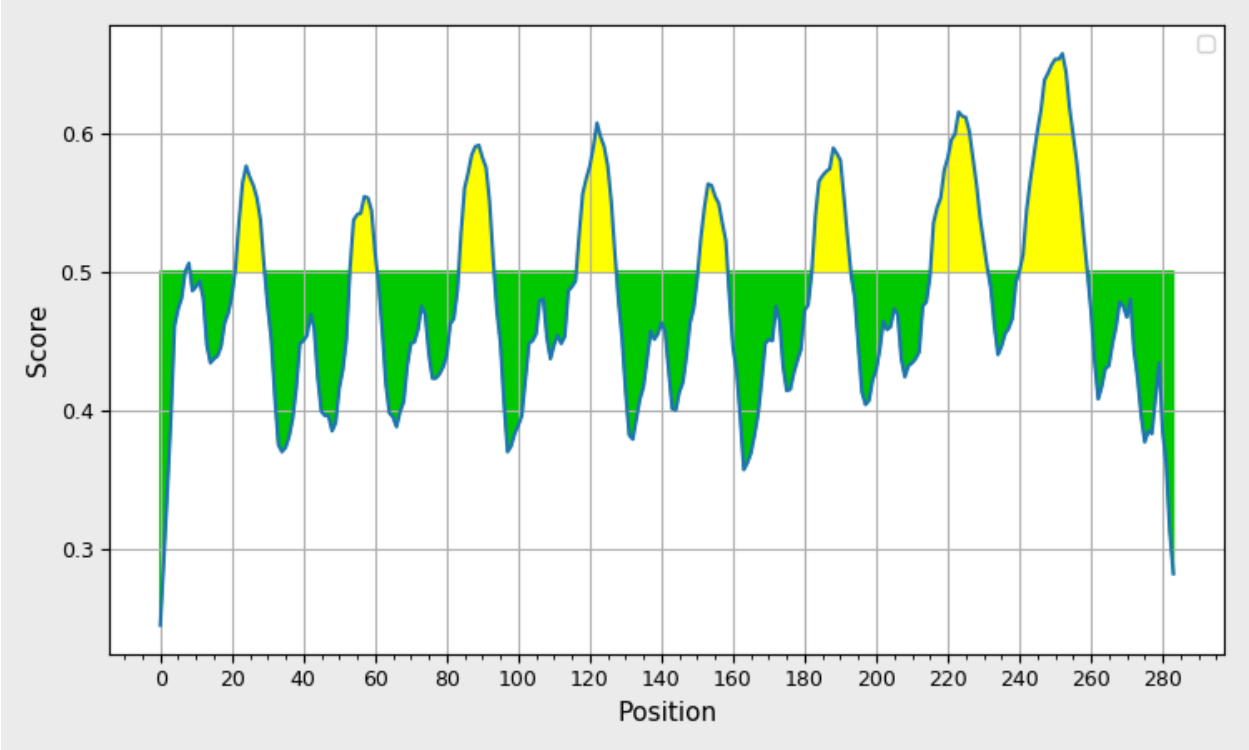
IYSVNLENVLLDDAEIAKMIIEKHVEYKSDSYTKDLDDVKNNKLDEIISKNEELRLMY

VNCVRKNY

Predicted peptides:

1	9	9	W	1
2	22	30	DAFKADING	9
3	54	61	ALKNLLDN	8
4	85	94	MDDSQFDDKG	10
5	118	128	WRLMFYGKTGW	11

6	152	159	PSPFDLAI	8
7	183	193	MTSTNTNSLLF	11
8	217	232	INIYSVNLENVLLDDA	16
9	241	259	KHVEYKSDSYTKDLDDV KN	19



Protein : 24

Input Sequences : MPXV|URK20463.1|C2L SPI-3 Serpin 1,2,3

>MIALLILSLACSVSAYRLQGFINAGILAYKNIQNGDEDDNDNIV

FSPFGYSFSMFMSLLSASGNTRVELLNTMNLRKRDLGPAFTELISGLAKMKTSKYTYT

DLTYQSFVDSTVCIKPSYYQQYHRFGLYRLNFRRDAVNKINSIVERRSGMSNVVDSTM

LDDNTLWAIINTIYFKGTWQYPFDITKTHNASFTNKYGTKTVPMMNVVTKLQGNTITI

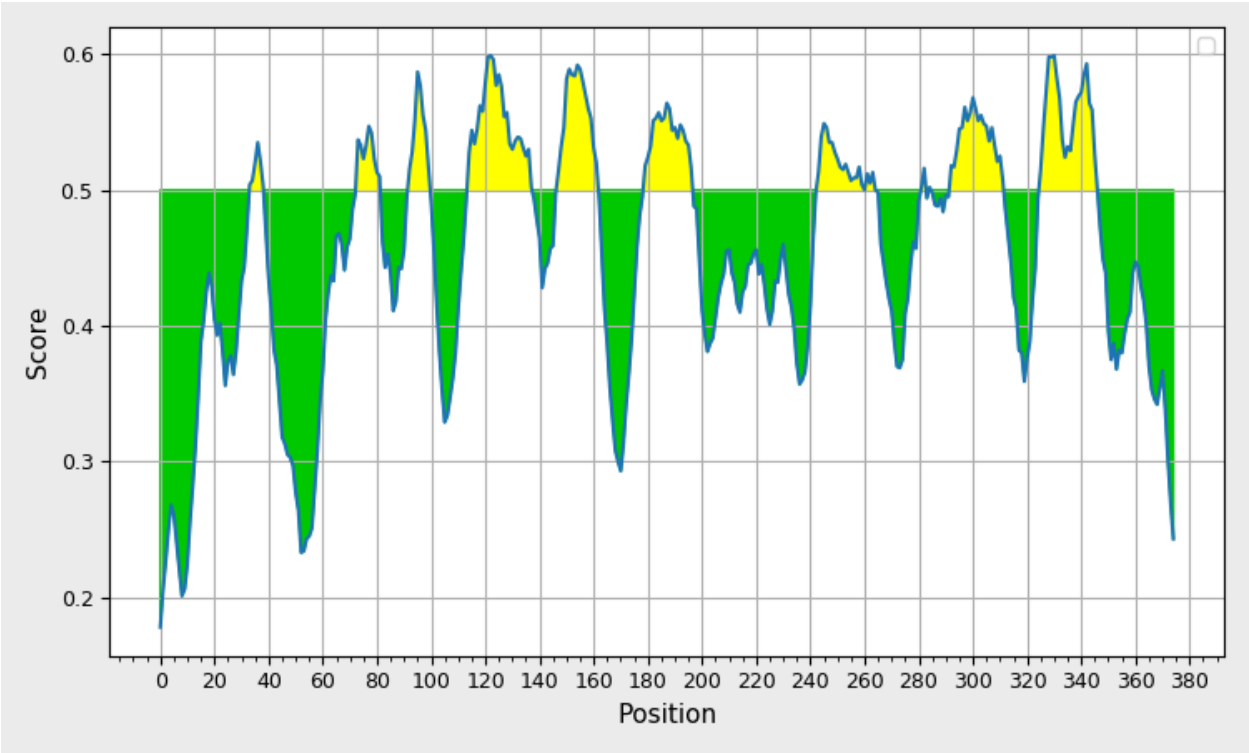
DDEEYDMVRLLYKDANISMYLAIGDNMTHFTDSITPAKLEYWSSQLGNKMYNLKLPRF

SIENKRDIKSIAEMMAPSMFNPDKASFKHMTRDPLYIYKMFQNVKIDVDEQGTVAEAS

TIMVATARSSPEELEFNTPFVFIIRHDITGFILFMGKVEYT

Predicted peptides:

1	34	39	NGDEDD	6
2	74	82	NLRKRDLGP	9
3	93	100	KMKTSKYT	8
4	115	138	CIKPSYYQQYHRFGLYRLNFRRDA	24
5	148	162	RRSGMSNVVDSTMLD	15
6	180	197	QYPFDITKTHNASFTNKY	18
7	244	260	NMTHFTDSITPAKLEYW	17
8	262	264	SQL	3
9	282	283	RD	2
10	285	285	K	1
11	293	312	PSMFNPDKASFKHMTRDPLY	20
12	326	346	EQGTVAEASTIMVATARSSPE	21



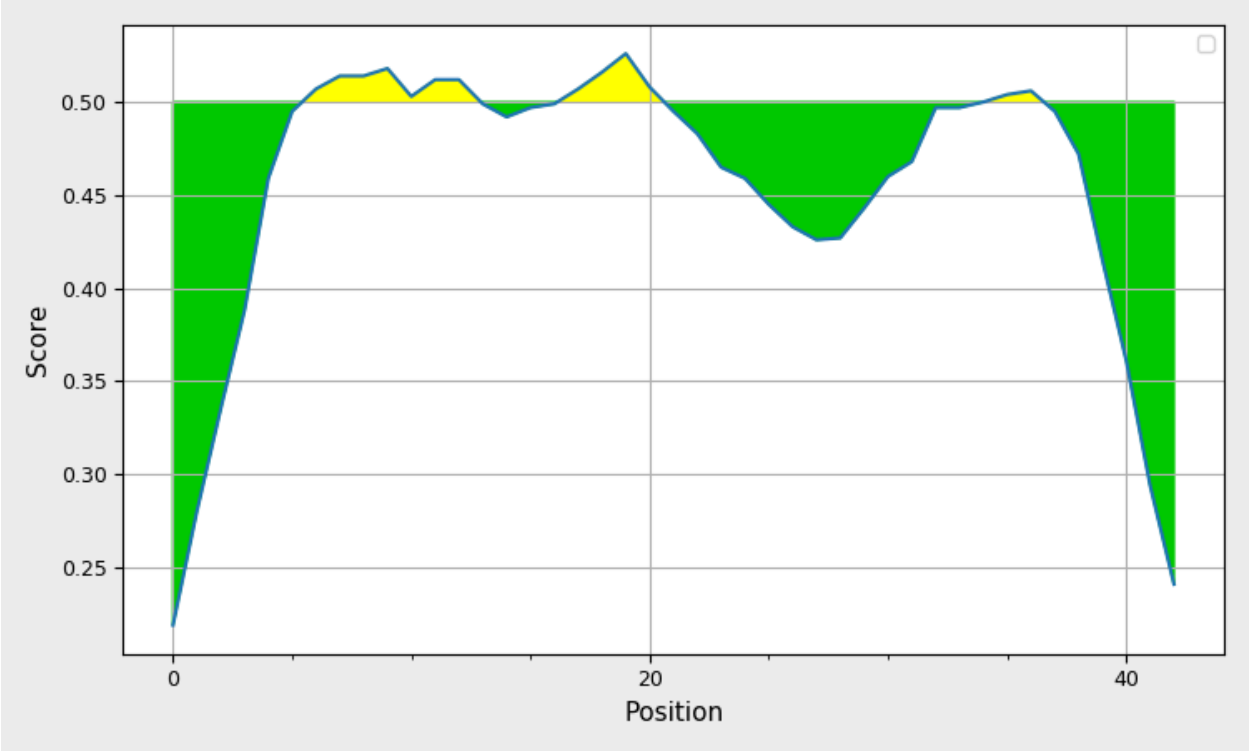
Protein : 25

Input Sequences : MPXV|URK20464.1|IFN resistance, PKR/eIF-alpha inhibitor

>MLAFCYSLPNAGDVIKGRVYENDYTLYIYLFDYPHFEAILAEC

Predicted peptides:

1	7	13	SLPNAGD	7
2	18	21	RVYE	4
3	36	37	FE	2



Protein : 26

Input Sequences : MPXV|URK20465.1|C4L Phospholipase-D-like protein

>MNPDNTIAVITETIPIGMQFDKVYLTTFNMWREILSNTIKTLDI

SSFYWSLSDEVGTNFGTVILNEIVQLPKRGVRVRVAVNKSNKPLKDVERLQMAGVEVR

YIDITNILGGVLHTKFWISDNTHIYLGSANMDWRSLTQVKELGIAIFNNRNLAADLTQ

IFEVYWYLGVNNLPYNWKNFYPSYYNTDHPLSINVSGVPHSVFIASAPQQLCTMERTN

DLTALLSCIGNASKFVYVSVMNFIPIIYSKAGKILFWPYIEDELRRSAIDRQVSVKLL

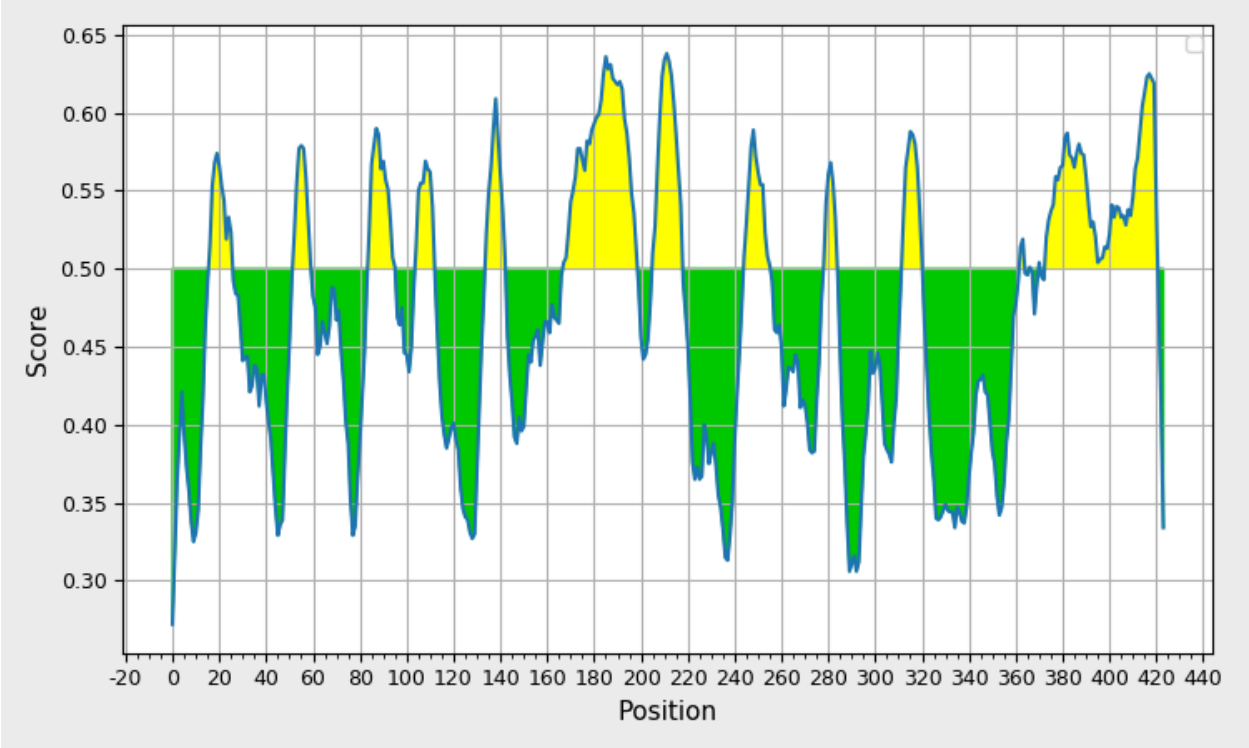
ISCWQRSSFIMRNFLRSIAMLKSKNIDIEVKLFIVPDADPPIPYSRVNHAKYMVTDKT

AYIGTSNWTGNYFTDICGASINITPDDGLGLRQQLEDIFIRDWNSKYSYELYDTSPTK

RCRLLKNMKQCTNDIYCDEIQPEKELPEYSLE

Predicted peptides:

1	17	26	GMQFDKVYLT	10
2	53	60	DEVGTNFG	8
3	85	96	NKPLKDVERLQM	12
4	105	112	DITNILGG	8
5	135	143	WRSLTQVKE	9
6	168	199	LGVNNLPYNWKNFYPSYYNTDHPLSINVSGVP	32
7	206	218	SAPQQLCTMERTN	13
8	245	256	IYSKAGKILFWP	12
9	279	284	CWQRSS	6
10	313	321	DADPPIPYS	9
11	363	364	LG	2
12	367	367	Q	1
13	371	371	D	1
14	374	421	IRDWNSKYSYELYDTSPTKRCRLLKNMKQCTNDIYCDEIQPEKELPEY	48



Protein : 27

Input Sequences : MPXV|URK20466.1|C5L Monoglyceride lipase

>MSTNCMFNLDNDYIYCKYWKPITYPKALVFISHGAGEHSGRYDE

LAENISSLGILVFSHDHIGHGRSNGEKMMIDDFGKYVRDVVQHVVTIKSTYPGVPVFL

LGHSMGATISILAACDNPNLFTAMILMSPLVNADAVPRLNLLAAKLMGTITPNASVGK

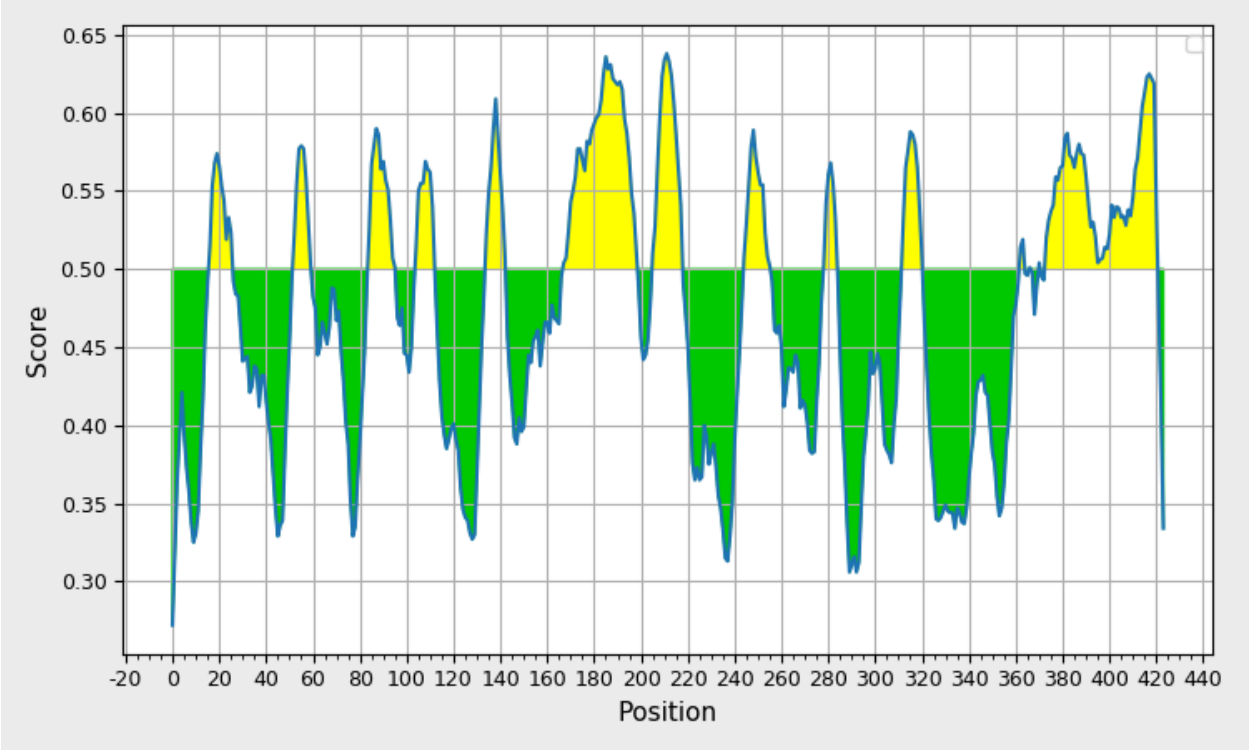
LCPESVSRDMDEVYKYQYDPLVNHEKIKAGFASQVLKATNKVRKIIPKINTPTLILQG

TNNEISDVSGAYYFMQHANCNREIKIYEGAKHHLHKETDEVKKSVMKEIETWIFNRV

Predicted peptides:

1	17	26	GMQFDKVYLT	10
2	53	60	DEVGTNFG	8
3	85	96	NKPLKDVERLQM	12
4	105	112	DITNILGG	8
5	135	143	WRSLTQVKE	9
6	168	199	LGVNNLPYNWKNFYPSYYNTDHPLSINVSGVP	32
7	206	218	SAPQQLCTMERTN	13

8	245	256	IYSKAGKILFWP	12
9	279	284	CWQRSS	6
10	313	321	DADPPIPYS	9
11	363	364	LG	2
12	367	367	Q	1
13	371	371	D	1
14	374	421	IRDWNSKYSYELYDTSPTKRCRLLKNMKQCTNDIYCDEIQPEKE LPEY	48



Protein : 28

Input Sequences : MPXV|URK20467.1|C6R Host immune response repressor

>METKSDYEDAVFYFVDDDEICSRDSIIDLIDEYITWRNHVIVFN

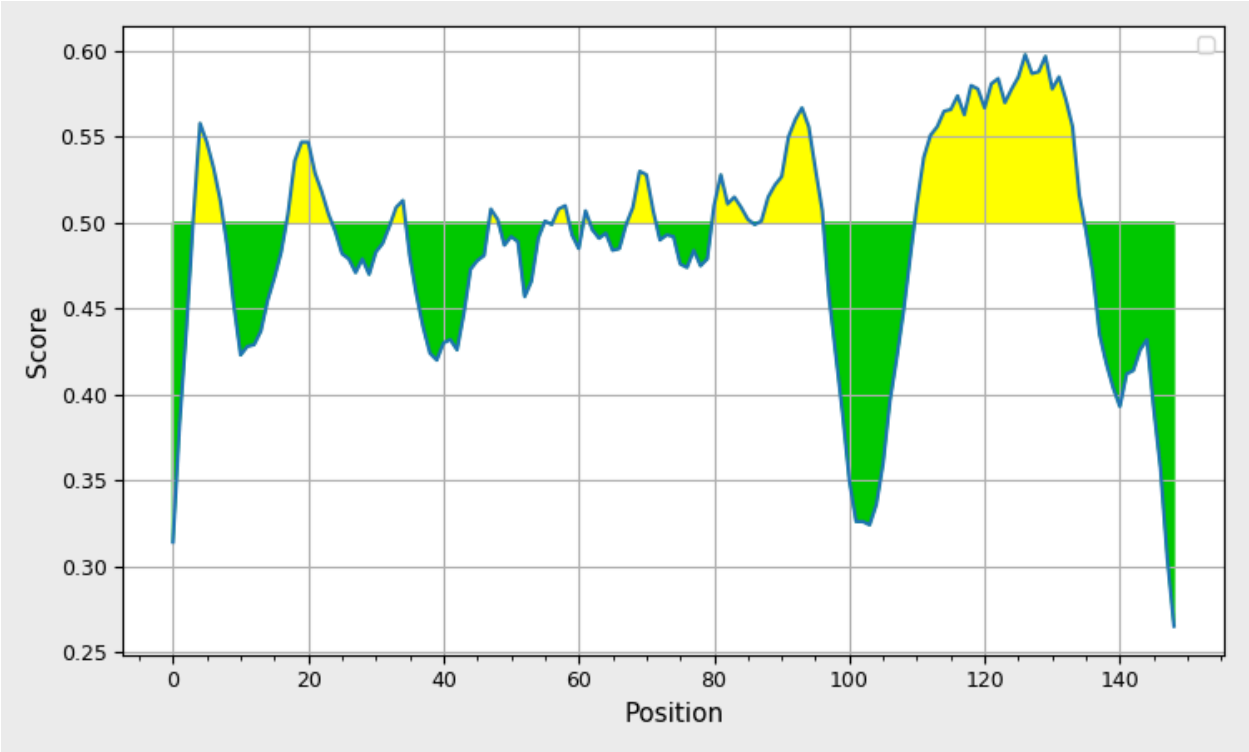
KDITSCGRLYKELIKFDDVAIRYYGIDKINEIVEAMSEGDHYINLTEVHDQESLFATI

GICAKITEHWGYKKISESKFQSLGNITDLMTDDNINILILFLEKKMN

Predicted peptides:

1	4	8	KSDYE	5

2	18	24	DEICSRD	7
3	34	35	IT	2
4	48	49	TS	2
5	56	56	E	1
6	58	59	IK	2
7	62	62	D	1
8	69	72	GIDK	4
9	81	86	SEGDHY	6
10	88	97	NLTEVHDQES	10
11	111	135	HWGYKKISESKFQSLGNITDLMTDD	25



Protein : 29

Input Sequences : MPXV|URK20468.1|C7L Caspase-9 (apoptosis) inhibitor (mitochondrial-associated)

>MLSMFMYNNIIDYVHVHDIEDEASDNDDRDYVYPLPENMVYRFD

KSTNILDYLSTERDHVMMAVQYYMSKQRLDDLYRQLPTKTRSYVDIINTYCDKVNNDY

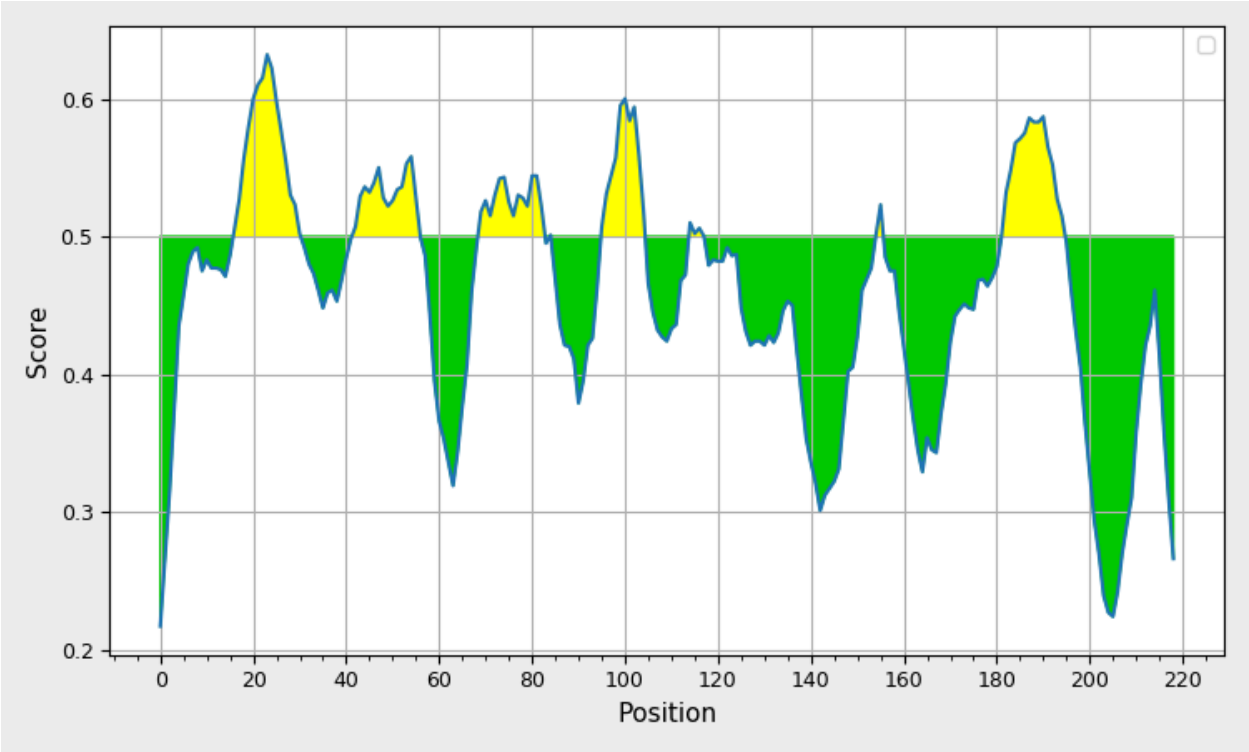
NSDMNIMCDMASTESFTVYDINNEVNTILMNNKGLGVRLATISFITELGRRCMNPVET

IKMFTLLSHTICDDYFVDYITYISAPRDNAIHSTREYLKIMGIAVIMFATYKTLKYMI

G

Predicted peptides:

1	17	31	HDIEDEASDNDDRDY	15
2	43	56	FDKSTNILDYLSTE	14
3	70	83	KQRLDDLYRQLPTK	14
4	85	85	R	1
5	96	105	DKVNNDYNSD	10
6	115	117	TES	3
7	155	156	MN	2
8	183	195	ISAPRDNAIHSTR	13



Protein : 30

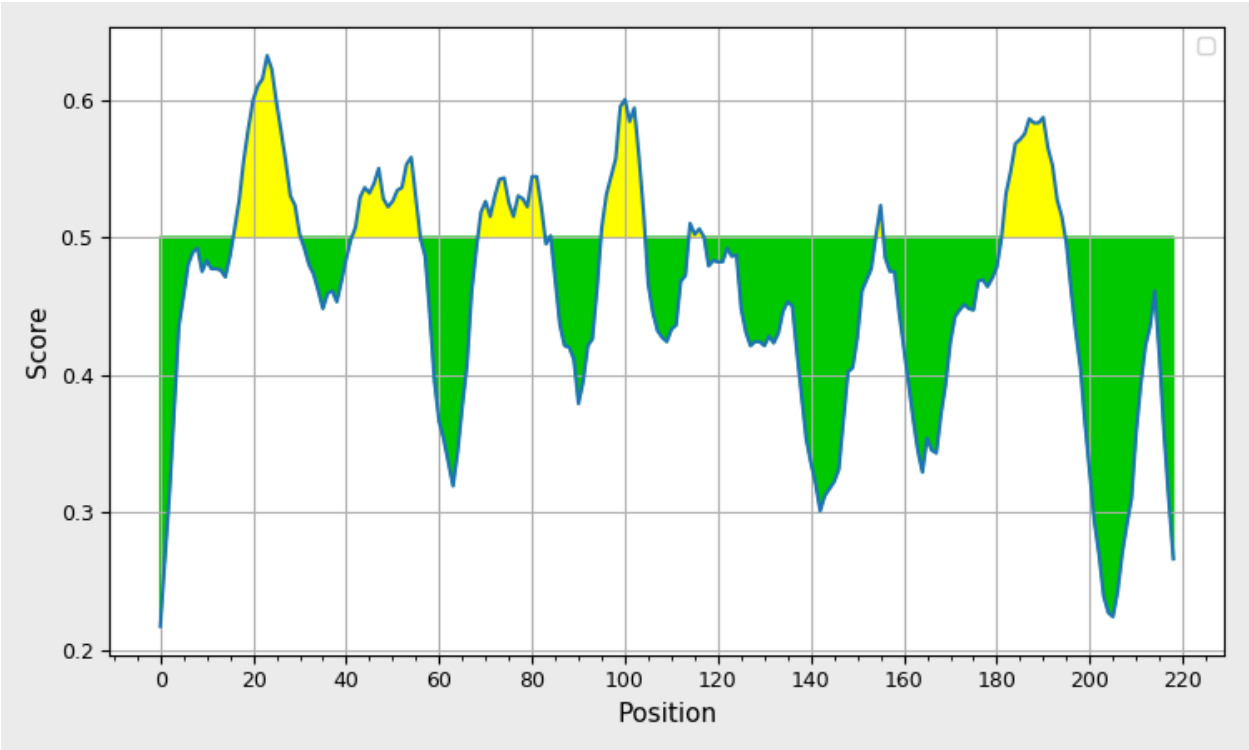
Input Sequences : MPXV|URK20469.1|C8L dUTPase

>MIFFMFNMNINSPVRFVKETNRAKSPTRQSPGAAGYDLYSAYDY

TIPPGERQLIKTDISMSMPKFCYGRIAPRSGLSLKGIDIGGGVIDEDYRGSIGVILIN

NGKCTFNVNTGDRIAQLIYQRIYYPELEEVQSLDSTDRGDQGFGSTGLR

1	17	31	HDIEDEASDNDDRDY	15
2	43	56	FDKSTNILDYLSTE	14
3	70	83	KQRLDDLYRQLPTK	14
4	85	85	R	1
5	96	105	DKVNNDYNSD	10
6	115	117	TES	3
7	155	156	MN	2
8	183	195	ISAPRDNAIHSTR	13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